

Ben Bubnick

Master Career History

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Career Overview

Interdisciplinary data engineer, analyst, software developer, and educator with a foundation in physics and fine arts and more than a decade of experience building reliable data systems in healthcare, insurance, and color science. Leads ambiguous, high-stakes work from source discovery and requirements through modeling, automation, validation, migration, documentation, and production handoff, spanning cloud data platforms, more than 100 billion clinical records, multi-organization EHR integrations, revenue-cycle models, and applied machine learning. Known for turning complex technical and scientific problems into reproducible systems, clear explanations, and measurable delivery improvements.

Career experience also includes college physics and astronomy teaching, published astrophysics and printmaking research, independent product development and consulting, visual arts, community service, organizational leadership, and hands-on farm and operations work. Across these settings, combines analytical rigor, creative problem solving, practical craftsmanship, continuous learning, and a commitment to making knowledge, tools, and opportunities more accessible to others.

Skills and Domains

Programming and Query Languages: *Current/Core:* SQL, Python, Ruby, Bash/Zsh; *Working Knowledge:* JavaScript, C#, Swift, Java, R; *Historical:* VB.NET, C++, FORTRAN, Perl, IDL, MQL, XPath, XQuery, JSONPath, Jaql

Application Development: *Working Knowledge:* APIs and web services, JSON and schema validation, document extraction and OCR-to-LLM parsing, native macOS development, SwiftUI, application permissions and accessibility configuration, scripted packaging, Spring Boot/Jersey; *Historical:* ASP.NET, MVC, Web APIs, React, Node.js, HTML5, CSS, LINQ, commercial-insurance rating applications

Databases and Query Engines: *Current/Core:* Snowflake, PostgreSQL, Amazon Athena, Amazon Redshift; *Working Knowledge:* Azure Synapse Analytics, Microsoft SQL Server and T-SQL, MySQL, MongoDB/PyMongo, SQL query tuning and indexing, partitioning and clustering; *Historical:* Oracle and PL/SQL, Hive, Impala, Greenplum

Data Architecture and Engineering: *Current/Core:* dbt, SQLMesh, Jinja-templated SQL, Snowpark, ETL/ELT, data integration and transformation, data warehouses and lakes, greenfield data-platform architecture, proof-of-concept development and productionization, source-system discovery and feasibility analysis, source-to-target mapping, dimensional modeling, business-grain definition, natural/composite/surrogate keys, identity crosswalks, metadata-driven transformations, data contracts and lineage; *Working Knowledge:* Databricks Lakehouse, Lakeflow Connect, Lakeflow Spark Declarative Pipelines, Spark, Kafka, Data Vault modeling, Parquet, OMOP Common Data Model; *Historical:* Hadoop/HDFS, Pig

Data Quality, Governance, and Migration: *Current/Core:* data-quality strategy and automated testing, acceptance criteria, null/duplicate/referential-integrity and business-rule checks, source-to-target and aggregate reconciliation, schema-drift detection, parity checks, regression testing, UAT, migration-acceptance testing, deidentification validation, root-cause analysis and defect remediation, idempotent and deterministic reruns, audit logging; *Working Knowledge:* Databricks data-governance fundamentals, Unity Catalog interoperability, catalog management and data organization, data privacy, governance at scale, role-based access control (RBAC) design, terminology mapping, data-sharing and consent constraints

Data Operations and Reliability: *Current/Core:* job scheduling, workflow orchestration, pipeline and software dependency management, observability, monitoring and alerting, event-driven workflow notifications, incident classification and response, production troubleshooting, runbooks, release and promotion gates, configuration and version management; *Working Knowledge:* Lakeflow Jobs, Databricks workload deployment and performance optimization

Cloud, DevOps, and Test Tools: *Current/Core:* AWS (S3, Glue, Lambda, SNS, event notifications, CloudWatch, AWS CLI), Microsoft Azure (Data Factory, Functions, Blob Storage), Git/GitHub, GitHub Actions, Docker, CI/CD, automated unit and integration testing, test fixtures and golden files, RSpec, RuboCop, Liquibase, Make, CLI environment setup, Python virtual environments; *Working Knowledge:* Databricks declarative automation bundles and data-engineering DevOps patterns, hermetic tests, Azure DevOps, Jenkins, Helm/Kubernetes, Gradle, Postman, regular expressions, Tesseract OCR, RubyLLM; *Historical:* cron

Healthcare Data, Interoperability, and Risk Adjustment: *Current/Core:* clinical, payer, eligibility, claims, revenue-cycle, and financial data; EHR/EMR integration; HL7 v2/v3, FHIR, EDI/837, electronic visit verification (EVV); patient/provider identity, population health, care gaps, HEDIS and other quality measures and registries, prior authorization, health-information exchange, ADT notifications, PHI handling and HIPAA-compliant deidentification workflows; Medicare risk adjustment and risk capture, RAF/HCC workflows, ICD-10-to-HCC mapping, diagnosis-submission lineage, chart-chase data preparation, MMR/MOR/MAO-004 and payer/CMS response reconciliation; *Working Knowledge:* social determinants of health, provider directories, patient consent, HITRUST readiness support

Analytics, Machine Learning, and BI: *Current/Core:* data analysis, statistics, machine learning, predictive modeling, pandas, Jupyter, Power BI, data visualization, dashboard and report design, BI visual standards, operational and executive reporting; *Working Knowledge:* model monitoring, Tableau, R/Shiny, time-series forecasting and ARIMA diagnostics, optimization, NLP and topic modeling, XGBoost, random forests, neural networks, classification and regression evaluation (confusion matrices, ROC/AUC, calibration); *Historical:* SAS/GRAPH

Scientific Research and Computing: *Current/Core:* reproducible analysis and research-data workflows; *Working Knowledge:* experimental design, measurement uncertainty, instrument calibration, numerical analysis, linear algebra, mathematical modeling and simulation, finite-difference methods, interpolation, numerical integration and differentiation, error and stability analysis; *Historical:* physics and astrophysics research, telescope observing plans and logs, laboratory equipment maintenance, infrared spectroscopy and photometry, astrometry and source cross-matching, stellar classification and isochrone comparison, optical measurement and spectrophotometry, color matching and formulation, radiative-transfer modeling, statistical risk analysis, Fourier analysis, Monte Carlo methods, IRAF, SpeXtool, DAOPHOT, PASCO DataStudio, Mathematica, MATLAB, LabVIEW, Maple, PDL, JMP

Teaching and Learning: *Current/Core:* technical and scientific writing, documentation, stakeholder communication, mentoring, public speaking, training, and translation of complex technical and quantitative concepts for varied audiences; *Working Knowledge:* curriculum and instructional design, recorded/online instruction, rubric design, learning-management systems, accommodations and accessibility practices (presentation, response, timing, and setting), formal evaluation of learning outcomes, research publication; *Historical:* physics and astronomy instruction, laboratory instruction, tutoring and test preparation, peer instruction, Just-in-Time Teaching, Interactive Lecture Demonstrations, student feedback-based instructional improvement, order-of-magnitude estimation and comparative-risk communication

Consulting, Leadership, and Business: *Current/Core:* requirements facilitation, client discovery, stakeholder alignment, cross-functional delivery, technical specifications, code review, team mentoring and onboarding, proposal writing, pricing, invoicing, contract negotiation, bookkeeping, accounting, tax compliance, marketing, client acquisition, promotional-material development; *Working Knowledge:* product prototyping, cost and throughput tracking, change management, intellectual-property portfolio support

Customer Service, Sales, and Administration: *Working Knowledge:* customer relationship management (CRM), conflict de-escalation, copyediting to a style guide; *Historical:* consultative selling, telephone outreach, point-of-sale transactions, cash reconciliation, returns processing, merchandising, inventory counts, library reference service and collection inventory, Library of Congress classification, editorial writing and community journalism, club administration and event planning

Community and Civic Service: *Working Knowledge:* volunteer scheduling, community-health data preparation; *Historical:* public-record data transformation, community meal distribution and guest service, youth mentoring and outreach

Creative and Visual Arts: *Historical:* relief, intaglio, screen, and monoprint processes; plate and screen preparation, proofing, editioning and print exchanges; painting, drawing, photography and papermaking; exhibition preparation, commissioned work, studio collaboration, process experimentation and safer-material substitution; Adobe Illustrator, Adobe Photoshop, GIMP, AutoCAD

Farm, Property, Logistics, and Industrial Operations: *Current/Core:* orchard establishment, pruning, irrigation, pest monitoring, harvest planning, seasonal tree care, grounds maintenance, supplies and equipment management; *Historical:* estimating and quoting maintenance jobs, surface preparation and painting, concrete repair, ladders and power tools, construction cleanup, warehouse picking/packing and loading, production-line preparation and shop-floor safety, scrap sorting, event setup/teardown, waste and recycling handling

Food Service and Hospitality: *Historical:* food preparation, meal-service coordination, coffee and espresso preparation, counter service, dish-room and work-area sanitation, stocking, opening and closing duties, Person in Charge (PIC) certification

General Tools: *Current/Core:* Microsoft Office, LaTeX; *Working Knowledge:* Jira, Confluence, Salesforce CRM

Professional Experience

HHAEExchange

Oct. 2025 - Jun. 2026

Senior Data Engineer

Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox

Brought in to unify two distinct data environments following the HHAEExchange-Sandata merger. Built shared Snowflake/dbt models and onboarding patterns, then identified and remediated data-quality defects affecting hundreds of millions of EVV records. Led healthcare source analysis, data-quality investigation and remediation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use. Contract via Health Data Movers.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, establishing reusable patterns for faster client onboarding.
 - S** Following the HHAEExchange-Sandata merger, reporting depended on two organizations and two source configurations whose entities, keys, data types, and lifecycle rules did not line up.
 - T** I was asked to create a shared warehouse and analytics pattern without disrupting the established HHA reporting path.
 - A** I inventoried the source structures, aligned 55 staging, core, and BI tables, preserved application identifiers alongside stable warehouse keys, normalized field types, and used paired source-specific CTEs plus a common source-system tag so both implementations could feed the same downstream contract.
 - R** The two environments could be consumed through one consistent set of Snowflake/dbt models, and the repeatable structure gave later source and client onboarding a proven starting point instead of another one-off build.
- Diagnosed and remediated a multi-payer EVV dashboard data-loss defect by tracing fact-to-dimension joins across 1.12 billion visit records; quantified 214.5 million records (19.2%) excluded by missing patient, service-code, and payer mappings, isolated 99.2% of in-scope missing-patient visits to the CareSource feed, and corrected the underlying integration defects.
 - S** A multi-payer EVV dashboard was undercounting visits, and it was unclear whether the loss came from dashboard filters or from the warehouse joins.
 - T** I needed to quantify the loss, identify the responsible dimension, and isolate the affected feed before changing production logic.
 - A** I started with 1,117,470,531 fact rows, measured survival after each inner join, and showed that only 903,000,825 rows survived: 214,469,706 visits, or 19.2%, were excluded. I then scoped the analysis to the dashboard's payer and one-year window, traced missing patient matches by payer and source, and found that CareSource represented 757,897 of 764,073 missing-patient visits, or 99.2%. I corrected the patient, service-code, and payer integration defects.
 - R** The dashboard issue had a measured root cause and a targeted remediation rather than an unverified filter change.
- Executed structural, row-level, transformation, and completeness QA for a 156-field Snowflake/dbt EVV visit model; uncovered and corrected defects associated with a 52.23% source-to-target row-count gap, 73.6% null patient identifiers, and 105.6 million orphaned visit relationships.
 - S** A new Sandata EVV visit model with 156 output fields was being prepared for the shared analytics layer, but a successful dbt run did not prove that the transformation preserved the source population or relationships.
 - T** I needed to validate its structure, row grain, transformations, and completeness before it could be trusted downstream.
 - A** I compared the schema and every output field, tested visit-key uniqueness, reconciled source and target counts, measured null distributions, validated deletion logic, and joined the target back to the patient model. That exposed a 221,074,458-to-105,617,693 row gap of 52.23%, 73.6% null patient identifiers, and 105,601,156 orphaned visits. I traced the failures to specific joins and mappings and corrected the associated defects.
 - R** The model moved forward with quantified risks, reproducible QA, and corrected logic rather than relying on superficial row samples.
- Validated and remediated an EVV patient-address model, resolving six documented defects involving deletion handling, multi-address coverage, usage-tag logic, and production data types; verified 1.31 million unique keys, zero missing parent references, and exact 158,084-row parity for the deduplicated parent-matched population.
 - S** The initial EVV patient-address model did not cover multiple addresses correctly, omitted deletion and usage-tag behavior, and later produced a production type failure when its source changed.
 - T** I needed to remediate the model and prove both its logical and production correctness.
 - A** I documented six issues, changed the model to reuse the patient entity, incorporated the appropriate supplemental and address sources, added deletion and usage-tag logic, aligned the key convention with HHA, and shipped a production hotfix for the type error. I then tested key uniqueness, parent references, mandatory-field nulls, join coverage, filters, and source-to-target parity.
 - R** All six issues were closed, 1,313,744 keys were unique, no parent references were missing, and the deduplicated parent-matched population reconciled exactly at 158,084 rows.

- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
 - S** Before committing to a unified EVV design, the team needed to know whether the Sandata snapshots contained the entities, history, keys, and relationships required to reproduce established analytics.
 - T** I was responsible for separating what could be modeled reliably from what required a different scope or an explicit limitation.
 - A** I profiled raw tables, compared them with the existing HHA contracts, traced candidate joins, tested cardinality and completeness, examined delete and update behavior, and identified where source grain or type differences would affect implementation. I used those findings to choose staging boundaries, source-specific CTEs, identity rules, and validation criteria before full development.
 - R** The delivery plan was grounded in actual source behavior, reducing avoidable redesign and making known constraints visible to stakeholders early.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.
 - S** Many requirements arrived as broad requests to make Sandata data behave like existing HHA analytics, while the intended grain, join behavior, lifecycle rules, and acceptable variance were not always explicit.
 - T** I needed to turn that ambiguity into instructions that engineers, reviewers, and downstream users could evaluate the same way.
 - A** I documented source-to-target mappings and acceptance criteria for schema, key uniqueness, referential integrity, row parity, null distributions, filters, and derived fields. I organized those methods into 12 investigation runbooks with executable queries, expected relationships, issue logs, severity, remediation, status, and sign-off sections.
 - R** Reviews moved from subjective checks to repeatable evidence, and the same documentation became an onboarding standard for diagnosing later data-quality problems.
- Defined surrogate-key, natural-key, and crosswalk strategies supporting stable identity resolution across related source systems.
 - S** HHA and Sandata reused local numeric identifiers across accounts, payers, and environments, so a patient, provider, office, service code, or authorization ID was not globally meaningful by itself.
 - T** I needed stable keys for shared facts and dimensions while retaining enough lineage to trace a record back to its source.
 - A** I identified the business grain for each entity, formed composite natural keys from the necessary account and domain identifiers, retained application IDs, mapped them to warehouse-level identifiers, and propagated those rules through parent-child relationships and crosswalks. I then used uniqueness and referential-integrity checks to expose collisions and missing links.
 - R** Related records could be resolved consistently across source systems, and later corrections to key logic could be made centrally without losing source traceability.
- Structured target models using FHIR-like concepts where appropriate to improve consistency and extensibility for downstream integration and analytics.
 - S** The merged environment represented related healthcare concepts through source-specific tables and names, which made it difficult for downstream consumers to reason consistently about patients, caregivers, organizations, locations, addresses, and their relationships.
 - T** I needed a target structure that was understandable across systems and could evolve without claiming full FHIR conformance.
 - A** I used FHIR-like resource boundaries where they fit the data, separated core entities from relationships and supporting attributes, preserved source provenance, and aligned identifiers so the same conceptual entity had a consistent role across the warehouse. I validated the approach against the actual reporting fields and join paths rather than forcing the source into an abstract standard.
 - R** The models became easier to map, document, and extend for both analytics and future integration use cases.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains.
 - S** The HHA and Sandata EVV feeds represented the same business concepts differently: deletion and active-state flags varied, optional fields were inconsistently populated, date ranges used different conventions, and local keys only made sense inside a source account.
 - T** I needed to make those differences predictable before the data reached shared organization, office, caregiver, and patient models.
 - A** I built reusable SQL/Jinja patterns around explicit casts, CASE and COALESCE rules, deterministic row ranking, sentinel dates, lifecycle filters, and source-specific branches that converged on a common column contract. I kept application IDs and source labels visible so exceptions remained traceable.
 - R** The same transformation approach could be applied across domains, reducing special-case logic and making cross-source results easier to test, explain, and extend.

- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
 - S** A large set of new EVV models was moving through multiple environments, but failures such as duplicate keys, orphaned relationships, bad lifecycle states, and quoted-column behavior were not covered consistently.
 - T** I needed repeatable checks that would catch structural and business defects before promotion.
 - A** I created or redesigned 30 SQL/dbt tests, including custom generic tests for quoted identifiers, and combined them with model assertions for not-null and uniqueness rules, referential integrity, accepted values, deletion logic, and source-to-target counts. I reran the focused checks after each mapping or key change so fixes were regression-tested rather than merely spot-checked.
 - R** The integration gained an automated quality gate that made failures reproducible, localized defects earlier, and provided objective evidence for review and release decisions.
- Collaborated with client and internal stakeholders to resolve source ambiguity, refine requirements, and translate business needs into production-ready data assets.
 - S** Several EVV behaviors could not be inferred from column names alone, including which address source was authoritative, how usage tags should be populated, and which keys or lifecycle rules downstream reports expected.
 - T** I needed to obtain decisions from the right stakeholders and convert them into deployable logic.
 - A** I brought concrete record examples, join counts, and alternative mappings to client and internal review discussions, captured the decisions in model mappings and QA issue logs, incorporated peer-review feedback, and verified the revised behavior in Snowflake before promotion.
 - R** Ambiguous business questions became explicit technical rules, the resulting models were reviewable against shared acceptance criteria, and stakeholders had a documented explanation of why the production logic behaved as it did.
- Resolved data-quality and mapping issues through targeted test-case analysis and review of failing source and target records.
 - S** Integration failures appeared as symptoms in downstream tables, including missing records, duplicate service codes, orphaned relationships, and a production data-type error, while the actual defect could originate in raw data, staging logic, a join, or a cast.
 - T** I needed to isolate each failure quickly without rewriting working portions of the pipeline.
 - A** I built narrow test cohorts, compared the same keys across raw, staging, core, and BI layers, decomposed joins one at a time, inspected representative failing records, and used count and null deltas to confirm the responsible transformation. I changed only the affected filters, joins, deduplication, or type handling and reran the same test case as a regression check.
 - R** The defects were corrected with traceable evidence and focused changes that limited risk to the broader integration.
- Established reusable patterns for onboarding additional sources and extending integration workflows.
 - S** The merger integration would not be the last source added to the platform, so a solution built around hard-coded, table-by-table exceptions would create the same onboarding problem again.
 - T** I needed to leave an extension path that another engineer could follow.
 - A** I standardized the layered staging-to-core-to-BI pattern, source-specific CTE plus common-union structure, application and warehouse key fields, source labels, dbt tests, QA runbooks, and versioned Snowflake changes through Liquibase and GitHub workflows. I kept source differences localized so a new adapter could reuse the common contract and quality checks.
 - R** Additional sources could be onboarded by implementing a known set of mappings and tests rather than redesigning the downstream model, improving consistency and reducing integration risk.

GrowData Analytics

Sep. 2024 - Oct. 2025

Principal ML Engineer

Technologies: Python, Ruby, RubyLLM, RSpec, RuboCop, Tesseract OCR, LLM APIs, Docker, AWS, GitHub Actions

Founded GrowData Analytics to explore how AI led analytics could help small local farms make better operational decisions. The work exposed the more fundamental obstacle that the source data was trapped in inconsistent invoices. I built a preproduction document-to-data product with OCR, validation, audit trails, operational analytics, and a lightweight AWS deployment.

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
 - S** Small local farms had valuable purchasing and operating data trapped in handwritten and printed invoices whose layouts and image quality varied widely.
 - T** I needed to prove that a document-to-data product could produce useful records without sacrificing accuracy or auditability.
 - A** I assembled a stratified 500-invoice test suite across representative vendors, layouts, and quality levels; combined image preprocessing, OCR, language-model extraction, a versioned JSON schema, and arithmetic reconciliation; and compared outputs field by field while iterating on prompts and post-processing rules.
 - R** The preproduction MVP achieved 95% field-level accuracy and produced structured transaction records that retained enough context for review and correction.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.
 - S** High average extraction accuracy was not sufficient for an operational product because uncertain invoices and transient provider failures still needed safe handling.
 - T** I needed to maximize unattended processing while ensuring questionable records were reviewed rather than silently accepted.
 - A** I added automated tests and audit events, bounded retry controls for transient failures, validation and reconciliation signals, and confidence-based routing that separated clean records from invoices requiring human review. I tracked system failures independently from data-quality exceptions so each could be improved appropriately.
 - R** The MVP reached 88% straight-through processing while holding system failure below 1%, with an explicit review path for the remainder.
- Added schema checks, cross-field validation, reconciliation, idempotent reruns, fixtures, golden-file tests, linting, and release automation.
 - S** Changes to OCR, prompts, or post-processing could improve one invoice while breaking required fields, totals, or previously correct layouts.
 - T** I needed a release gate that protected the transaction contract and made reprocessing safe.
 - A** I enforced a versioned JSON schema, added cross-field and line-to-total reconciliation, designed idempotent reruns, built representative fixtures and golden-file regression tests, and connected tests and linting to release automation.
 - R** Changes could be evaluated against known invoices before release, malformed or inconsistent records were stopped early, and failed work could be rerun without creating duplicate transactions.
- Productized Tesseract OCR and rule-based post-processing as a small Python/Ruby service with containerized deployments and versioned configuration.
 - S** Early OCR experiments worked as isolated scripts but were sensitive to machine setup, image preparation, and unrecorded parameter changes.
 - T** I needed to turn that experimentation into a reproducible component that could be operated and improved like a product.
 - A** I wrapped Tesseract behind an extraction interface, added repeatable image orientation, grayscale, contrast, and normalization steps, applied rule-based cleanup, exposed the workflow through a small Python/Ruby service, and packaged its dependencies and versioned settings in containers.
 - R** The same invoice and configuration produced consistent behavior across environments, making OCR changes easier to test, deploy, and roll back.
- Built a lightweight analytics layer tracking usage, error categories, latency, and throughput to guide feature tuning and cost control.
 - S** Without operational measurements, I could not tell whether a feature improved customer value or merely added latency and model cost.
 - T** I needed enough observability to guide a small product without creating a large data platform.
 - A** I captured usage, latency, throughput, review routing, and categorized error events; summarized them by workflow and configuration version; and compared quality and processing behavior before and after changes.
 - R** The lightweight analytics layer exposed bottlenecks and recurring failure patterns, gave me evidence for feature and configuration tuning, and kept infrastructure and model-spend decisions tied to actual use.

- Led marketing and client acquisition, including creating promotional materials for the business.
 - S** A technically viable invoice parser still needed a clear value proposition and direct feedback from the small farms it was intended to serve.
 - T** I needed to explain the product in practical terms, create interest, and turn outreach into product learning and client opportunities.
 - A** I translated the workflow into benefits such as reduced data entry and better expense visibility, created promotional and demonstration materials, conducted outreach and product conversations, and used objections and requests to refine positioning and the roadmap.
 - R** GrowData established an initial acquisition process and a feedback loop connecting market conversations to product priorities.
- Managed infrastructure as code for a minimal AWS footprint using S3, Lambda or scheduled jobs, and CloudWatch alerts through Bash/Zsh and Makefile workflows.
 - S** As a self-funded preproduction product, GrowData needed reliable execution and monitoring without the cost or operational burden of a large platform.
 - T** I needed to deploy the pipeline in a minimal, reproducible AWS footprint.
 - A** I managed the infrastructure as code, used S3 for controlled document and result storage, selected Lambda or scheduled execution according to workload needs, configured least-privilege access and CloudWatch alerts, and wrapped common deployment and operating steps in Bash/Zsh and Makefile workflows.
 - R** The service could be recreated and monitored consistently, remained inexpensive at preproduction volume, and avoided manual environment configuration.
- Documented runbooks and failure modes covering timeouts, low-confidence OCR, and retries.
 - S** The pipeline depended on image quality, local OCR, and external model APIs, so failures ranged from slow requests to readable but low-confidence results.
 - T** I needed predictable recovery procedures that did not depend on remembering how a particular failure had been solved before.
 - A** I cataloged timeout, malformed-response, low-confidence OCR, validation, reconciliation, and exhausted-retry scenarios; documented the relevant logs and checks; and defined when to retry, preprocess again, route to review, reprocess, or escalate.
 - R** Incidents could be classified and handled consistently, limiting blind retries and reducing the chance that an uncertain invoice would enter the transaction dataset.
- Managed the business's bookkeeping, accounting, and tax obligations.
 - S** Building the product as a solo principal also made me responsible for the financial records and compliance work behind the business.
 - T** I needed to maintain an accurate view of cash activity and meet filing obligations while continuing product development.
 - A** I separated and categorized business income and expenses, reconciled accounts, retained source documents, maintained bookkeeping records, monitored upcoming obligations, and prepared the information needed for accounting and tax filings.
 - R** The business maintained organized, current records and met its administrative obligations without requiring a separate operations team.

Adonis

Mar. 2025 - Jul. 2025

Senior Data Engineer*Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker, GitHub Actions*

Joined a short-turnaround engagement to onboard Hopebridge Autism Therapy Center onto a revenue-cycle data platform. Transformed poorly documented source data into 54 tested Snowflake models, resolved fundamental key usage problems, and delivered a reusable onboarding and validation framework. Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding, analytics, and reporting. Contract via Veersa Technologies.

- Diagnosed undocumented source-key versioning and implemented composite business grains with surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
 - S** Practice Suite reused or regenerated identifiers as claims were rebilled and payments were posted, but that versioning behavior was not documented; identifiers that appeared unique at first produced duplicate-key failures when the full history was modeled.
 - T** I needed to establish the true business grain for charge and payment data without dropping legitimate activity.
 - A** I profiled the duplicate cohorts, traced claim numbers back to the stable accession number, and tested combinations of claim, line, charge, payment, payer, balance, and transaction attributes. I encoded the resulting composite grains and deterministic SHA-256 surrogate keys across 10 charge, payment, remittance, and transaction models, then propagated the keys through downstream relationships and reran uniqueness and reconciliation checks.
 - R** The changes eliminated the duplicate-key failures while preserving distinct rebills and payment events, giving downstream joins stable claim-, service-line-, and transaction-level identities.

- Built, documented, validated, and maintained 54 SQLMesh models that delivered standardized revenue-cycle datasets for a provider whose network served 4,598 families across 113 locations and processed 86,388 annual authorizations.
 - S** Hopebridge needed a new Practice Suite integration on a short timeline, but the source arrived as a poorly documented maze of database tables whose fields did not map cleanly to the Adonis revenue-cycle model.
 - T** As the engineer doing the main ETL and coding work, I needed to turn that raw source into a complete, supportable data product for claims, worklists, reporting, and downstream analytics.
 - A** I built, documented, tested, and maintained 54 SQLMesh models—32 intermediate transformations, 14 core entities, and eight intelligence aggregates—covering patients, providers, organizations, claims, service lines, transactions, remittances, and balances. I reconciled the outputs to source records and to workflows reviewed with the client and product team.
 - R** I delivered standardized Snowflake datasets for a network serving 4,598 families across 113 locations and processing 86,388 annual authorizations, with a structure another engineer could operate and extend.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
 - S** A successful SQLMesh plan only proved that the transformations executed; it did not prove that they preserved the intended population, respected each model's grain, or populated required fields.
 - T** I needed an objective release gate that would expose data defects before Hopebridge used the integration in production workflows.
 - A** I attached reusable uniqueness, not-null, composite-key, and source-to-target row-count audits to the Practice Suite models, including a threshold-based comparison audit for complex transformations. I also created a model-by-model QA runbook covering field population, referential integrity, allowed values, dates, and representative business rules, then investigated failures and reran the same checks after each correction.
 - R** At final delivery, all 117+ configured audit applications passed, providing reproducible evidence that the 54-model integration met its structural and completeness expectations.
- Built a reusable SQLMesh/Snowflake ingestion and transformation framework using templated models, data contracts, CI checks, and promotion gates.
 - S** The broader integration repository was moving from legacy, client-specific implementations toward a shared SQLMesh structure, while Practice Suite still required source-specific mappings and client schemas.
 - T** I needed to implement Hopebridge in a way that fit the common framework and could be reused instead of becoming another isolated pipeline.
 - A** I organized the work into intermediate, core-entity, and intelligence layers; parameterized database and client schemas; reused macros for identifiers and aggregates; declared grains, column contracts, and audits with the models; and exercised SQLMesh plans in isolated environments before promotion. I also reviewed patterns used by other integrations and fed implementation lessons back into the shared approach.
 - R** Practice Suite became a repeatable SQLMesh/Snowflake project with source-specific logic localized behind a consistent contract, allowing safer CI checks and promotion while giving later client work a practical reference design.
- Standardized transformation logic across clients and translated business requirements into technical specifications to improve delivery predictability.
 - S** Important Practice Suite behavior was encoded in client workflows rather than the database documentation: collection status carried different meaning from claim status, rebills received new claim numbers, accession numbers remained stable, and requested fields such as taxonomy were not always in the delivered source.
 - T** I needed to convert those discoveries into precise implementation rules while keeping them compatible with shared Adonis data contracts.
 - A** In client and product working sessions, I used concrete records and screens to ask how rebills, submission dates, worklists, and identifiers were actually used. I translated the answers into model grains, mappings, filters, required-field gaps, audit criteria, and technical specifications, and compared the approach with peer integrations during review.
 - R** Business ambiguity became testable transformation logic, the team could identify missing vendor data early, and shared patterns improved delivery predictability without erasing the source-specific rules Hopebridge depended on.

- Authored implementation playbooks, technical handoff documentation, and a QA runbook for a repeatable healthcare-data validation workflow.
 - S** The engagement combined a compressed delivery schedule with a source system that required extensive discovery, so undocumented knowledge about setup, mappings, validation, and recovery would have made the integration difficult to support after handoff.
 - T** I needed to make both the implementation process and its quality checks repeatable for teammates working on Practice Suite and other integrations.
 - A** I documented the SQLMesh project setup, layered model conventions, key and audit patterns, plan and run workflow, and troubleshooting steps. I created a detailed QA runbook for 14 core subject areas with required audits, field-population checks, referential-integrity tests, business rules, and items to preserve for future maintainers, and I used the same structure during peer review and handoff.
 - R** The team received an executable validation and support playbook rather than tribal knowledge, improving the consistency of reviews, onboarding, and post-delivery troubleshooting.
- Added schema-drift alerts, parity checks, reconciliation logs, deterministic rerun procedures, and incident-handling guidance for UNIX-like environments.
 - S** Practice Suite data and SQLMesh plans could change independently, and a rerun after an upstream schema change or partial failure risked producing inconsistent results without a clear recovery path.
 - T** I needed to make the integration observable and safe to rerun in a UNIX-like operating environment.
 - A** I added schema and contract checks, source-to-target parity tests, and reconciliation logging around model execution; used deterministic composite and surrogate keys so the same inputs recreated the same identities; and documented how to inspect SQLMesh logs, isolate the failed model or audit, correct the cause, rerun in a development environment, and promote only after the checks passed. I coupled those controls with alerts for material drift and failed validation.
 - R** Schema changes and incidents became diagnosable events with a repeatable recovery sequence, reducing the risk that a partial rerun or silent drift would reach downstream reporting.
- Packaged a lightweight operations forecasting and alerting pattern as a tested Python and Docker service with GitHub Actions deployment automation.
 - S** The team needed earlier warning when operational volume or processing behavior moved outside expected ranges, but a heavyweight platform change was not appropriate for the engagement.
 - T** I needed to create a small, portable forecasting and alerting capability that could be tested and deployed consistently.
 - A** I implemented the pattern as a Python service that derived an expected operating range from historical observations, compared new results with that baseline, and emitted actionable alerts for material deviations or failed runs. I added automated tests for normal, threshold, and failure paths, packaged the service in Docker, and configured GitHub Actions to validate and deploy the same artifact.
 - R** The team gained a reusable, low-overhead monitoring component that converted operational variance into an early signal and could be promoted without environment-specific manual steps.

Arkos Health

Jun. 2023 - Jan. 2025

Senior Data Analyst

Technologies: AWS S3, Athena, Redshift, Glue, SNS, dbt, PySpark, Docker, Jupyter, Power BI, Tableau

Joined as a senior data analyst supporting Medicare risk capture, then grew into the data engineer and platform builder the team needed. Built a greenfield AWS data-lake proof of concept using Glue, Athena, event notifications, and SNS. After validating the approach, designed and built the final architecture spanning the data lake and Redshift, using dbt as the transformation and modeling layer.

- Architected and built a greenfield AWS data-lake foundation integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets; CI/CD automation and targeted model refactoring accelerated deployments and improved query performance.
 - S** Arkos needed to replace fragmented, source-specific reporting with a governed platform that could support six analysts and six state markets.
 - T** I was responsible for proving the architecture and turning it into a production data lake that could absorb 18 operational, clinical, and payer sources.
 - A** I designed the S3 raw, bronze, staging, silver, and gold layers; built Glue, Athena, Redshift, and dbt processing; standardized configuration and scheduling; and introduced CI/CD and targeted refactoring as the model estate grew.
 - R** The platform scaled to roughly 200 scheduled assets and 200 dbt models, gave analysts repeatable governed datasets, and made releases and performance improvements safer to deliver.

- Configured event notifications for the greenfield data-lake CI/CD workflows, routing workflow status and failure events through Amazon SNS for centralized tracking.
 - S** As automated deployments expanded, workflow outcomes were visible in individual AWS or CI/CD consoles but were not collected in one place.
 - T** I needed to make deployment and failure states observable without requiring analysts to continually inspect each workflow.
 - A** I wired workflow status and failure events into Amazon SNS, standardized the event details needed for triage, and validated that successful and failed executions produced usable notifications and audit records.
 - R** The team gained a central signal for CI/CD activity, could identify failed runs more quickly, and had the context needed to begin follow-up without first reconstructing the execution path.
- Established SNS topics and subscriptions for CI/CD workflow notifications, giving the analytics team timely visibility into deployment outcomes and supporting follow-up on failed runs.
 - S** Producing workflow events alone did not ensure that the right people would receive actionable information at the right time.
 - T** I needed to turn those events into a dependable notification channel for the analytics team.
 - A** I created and organized SNS topics and subscriptions for CI/CD outcomes, connected publishers and consumers, tested delivery and failure behavior, and documented how the team should use the notifications during support.
 - R** Analysts received timely deployment and failed-run visibility, reducing the chance that a broken workflow would go unnoticed and establishing a repeatable path from alert to investigation.
- Built and maintained the risk-adjustment SQL/dbt model inventory; supported three RAF notebooks, their companion workbook, and the Tableau Risk Dashboard while preserving an auditable record of workflow logic, inputs, outputs, and provenance.
 - S** Risk-adjustment logic was distributed across SQL/dbt models, three team-authored RAF notebooks, a companion workbook, and a Tableau dashboard, which made dependencies and ownership easy to blur.
 - T** I needed to make the backend model estate supportable while preserving accurate provenance.
 - A** I built and maintained the SQL/dbt inventory, documented inputs, outputs, dependencies, and business rules, and helped notebook, workbook, and dashboard owners reconcile their results to the governed models.
 - R** The team had an auditable chain from source data to analytical output, could diagnose mismatches more quickly, and retained a clear distinction between my model work and the presentation assets built by others.
- Built the SQL/dbt ingestion and normalization layer used by version-aware AWS Glue and Athena pipelines for evolving MAO, HCC, MOR, and related payer files, preserving source lineage and standardizing downstream datasets in partitioned Parquet across the lakehouse.
 - S** MAO, HCC, MOR, and payer files arrived in changing layouts and versions, which could break downstream risk models or silently shift field meaning.
 - T** I needed to absorb those variations while presenting stable analytical datasets.
 - A** I built source- and version-aware Glue and Athena ingestion, normalized the outputs through SQL/dbt contracts, preserved file and row lineage, handled latest-record logic, and wrote partitioned Parquet into governed lakehouse layers.
 - R** Downstream models consumed consistent structures despite upstream format changes, and analysts could trace normalized records back to the originating file and version.
- Built SQL/dbt models for multi-market Medicare RAF data preparation in Arizona, North Dakota, and Illinois, combining enrollment and MMR demographic factors with medical claims, signed EHR diagnoses, and payer encounter and MAO diagnoses; generated member and diagnosis inputs supporting CMS-HCC scoring and pre-/post-wellness-visit RAF analysis.
 - S** Arizona, North Dakota, and Illinois had different payer feeds and operational patterns, while RAF analysis required a consistent view of member eligibility, demographics, and diagnoses.
 - T** I needed to prepare reliable member and diagnosis inputs for CMS-HCC scoring and pre-/post-wellness analysis.
 - A** I built SQL/dbt models that joined enrollment and MMR factors to claims, signed EHR diagnoses, payer encounter data, and MAO results; aligned member, contract, and service-date keys; and reconciled market-specific exceptions.
 - R** The risk team received comparable multi-market inputs and could evaluate RAF movement using a documented data chain instead of separate ad hoc extracts.

- Built SQL/dbt models implementing diagnosis-submission lineage and acceptance logic covering add/delete activity, repeated submissions, encounter disposition, source and date-of-service preservation, and selection of the latest accepted diagnosis state.
 - S** A diagnosis could be submitted repeatedly, added or deleted, and receive different encounter dispositions over time, making a simple latest-row query unreliable.
 - T** I needed to determine the defensible current state without losing the history required for audit.
 - A** I defined the diagnosis grain and ordering rules, preserved source and date of service, modeled add/delete and repeated-submission activity, incorporated encounter disposition, and selected the latest accepted state while retaining lineage.
 - R** Analysts could explain why a diagnosis was counted or excluded and trace the answer back through each submission event.
- Built SQL/dbt reconciliation models that traced diagnoses from signed EHR encounters through Waystar claims, supplemental files, and MAO-004-style payer and CMS responses, enabling the risk team to isolate where diagnoses dropped from the submission path.
 - S** When a diagnosis appeared in a signed EHR encounter but not in a payer or CMS result, the risk team could not readily see where it left the submission chain.
 - T** I needed to make each handoff reconcilable.
 - A** I built SQL/dbt models that linked signed EHR diagnoses to Waystar claims, supplemental outputs, payer responses, and MAO-004-style CMS responses; standardized matching keys and assigned stage-specific statuses for missing and accepted records.
 - R** The risk team could isolate the point of failure, investigate the appropriate owner or file, and explain the final disposition with end-to-end lineage.
- Built SQL/dbt models for diagnosis-level HHS-HCC capture analytics that mapped ICD-10 diagnoses from signed face-to-face visits, reconciled them with payer gap reports, classified existing, new, and net-new capture, identified the first capture source, and produced monthly capture and HCC-per-visit measures.
 - S** Marketplace risk teams needed to know whether diagnoses from signed face-to-face visits actually closed payer-identified HHS-HCC gaps and which encounters created new value.
 - T** I needed to turn visit and gap data into diagnosis-level capture measures.
 - A** I mapped ICD-10 diagnoses to HCCs, reconciled signed visits with payer gap reports, classified existing, new, and net-new capture, identified the first capture source, and aggregated monthly capture and HCC-per-visit measures.
 - R** Stakeholders could distinguish activity from true incremental capture and evaluate performance with traceable diagnosis-level evidence.
- Built SQL/dbt models and datasets for supplemental diagnosis, claims, and member-enrollment workflows serving Allwell, NextBlue, Humana, Allymar, and Reveleer, joining claims to the appropriate member, MBI, enrollment period, contract, demographic segment, and member-month and formatting outputs to downstream specifications.
 - S** Allwell, NextBlue, Humana, Allymar, and Reveleer each required different supplemental diagnosis, claims, or enrollment outputs, but all depended on accurate member and coverage alignment.
 - T** I needed to make those deliveries repeatable without duplicating core business logic for every recipient.
 - A** I built shared SQL/dbt preparation models, joined claims to the correct member, MBI, enrollment period, contract, demographic segment, and member month, then applied recipient-specific formatting and validation rules.
 - R** The team could produce multiple downstream files from a consistent foundation, with clearer QA and less manual reassembly for each delivery.
- Built a dbt-based Reveleer chart-chase data product across three Medicare contracts, covering eligible-member and encounter selection, market-specific rules, provider and chart-location resolution, stable chase identifiers, and generation of separate chase and diagnosis files.
 - S** Reveleer chart-chase submissions spanned three Medicare contracts with different eligibility, encounter, provider, and chart-location rules.
 - T** I needed to produce compliant files while keeping every selected member and diagnosis traceable across markets.
 - A** I built a dbt model chain for eligible-member and encounter selection, encoded market-specific exclusions, resolved providers and chart locations, created stable chase identifiers, and generated separate chase and diagnosis outputs with validation checks.
 - R** Arkos had a repeatable multi-contract chart-chase product whose records could be reconciled from source data through the delivered files.

- Built multi-market SQL/dbt datasets combining member, diagnosis, MOR, payer-gap, and ICD-to-HCC mapping data to support analysis of the CMS-HCC V24-to-V28 transition.
 - S** The CMS-HCC transition from V24 to V28 changed diagnosis mappings and risk categories, so leadership needed to understand the effect across markets.
 - T** I needed to create a comparable analytical foundation for both model versions.
 - A** I built SQL/dbt datasets joining member, diagnosis, MOR, payer-gap, and ICD-to-HCC mapping data, calculated version-specific classifications, and reconciled records that moved, disappeared, or emerged under V28.
 - R** Analysts could evaluate the transition with consistent multi-market inputs, explain category-level changes, and focus follow-up on the members and diagnoses driving the differences.
- Integrated existing Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting.
 - S** Analysts already had valuable Jupyter-based work, but it depended on individual knowledge and was disconnected from the new governed lakehouse.
 - T** I needed to preserve that analytical value while making the platform understandable and usable without constant engineering assistance.
 - A** I integrated reusable notebook workflows with governed data, documented the platform architecture, environments, model layers, operating procedures, and lineage, and supplied examples for recurring, executive, and BI reporting.
 - R** Six analysts could independently find, query, and extend trusted datasets, which reduced one-off handoffs and made the new platform part of their normal reporting workflow.
- Instrumented scheduled pipelines with monitoring, alerting, and more than 10 failure classifications; authored three AWS/dbt incident runbooks supporting rapid detection and recovery.
 - S** A growing schedule of AWS and dbt jobs could fail for very different reasons, and raw error messages did not provide a consistent response.
 - T** I needed to shorten the time from a failed run to a correct recovery action.
 - A** I instrumented execution history, logs, data-freshness checks, and alerts; grouped recurring failures into more than 10 operational classifications; and wrote three incident runbooks covering detection, diagnosis, remediation, and escalation.
 - R** The team could distinguish source, infrastructure, transformation, and data-quality failures more quickly and recover scheduled pipelines through a repeatable process instead of ad hoc troubleshooting.
- Expanded cross-departmental collaboration to bring operational context into analytics decisions.
 - S** Technically correct models could still miss how risk, quality, clinical, operations, finance, and BI teams used the data in practice.
 - T** I needed to bring that operational context into design decisions before logic became embedded in production.
 - A** I held working sessions with stakeholders, brought sample records and reconciliation results into the discussion, translated business rules into explicit model logic, and documented decisions and exceptions for reviewers.
 - R** Models reflected real workflows, stakeholders could validate the logic in familiar terms, and downstream teams had greater confidence in the resulting measures and datasets.
- Initiated recurring stakeholder-feedback sessions to refine deliverables against business needs.
 - S** Requirements for risk and operational reporting continued to evolve after initial delivery, creating a risk of late rework.
 - T** I needed a reliable way to validate assumptions while the work was still inexpensive to change.
 - A** I initiated recurring feedback sessions, demonstrated prototypes and reconciliations, walked stakeholders through edge cases, captured decisions and open questions, and fed the results back into model and delivery priorities.
 - R** Misunderstandings surfaced earlier, deliverables stayed closer to current business needs, and stakeholders had a regular forum for shaping the work rather than waiting for a final handoff.
- Mentored engineers on code review, testing, and pipeline design and authored internal delivery standards.
 - S** As more contributors added pipelines and models, inconsistent grains, tests, naming, and review practices increased maintenance risk.
 - T** I needed to raise delivery consistency while helping engineers make sound design decisions themselves.
 - A** I coached teammates through code reviews, explained the reasoning behind test and modeling feedback, paired on pipeline design, and documented standards for naming, model contracts, validation, documentation, and pull-request readiness.
 - R** Reviews became more predictable, defects were caught earlier, and engineers had a shared set of expectations they could apply before requesting review.

- Led goal-setting sessions to establish shared objectives and a consistent tracking cadence.
 - S** The team was balancing platform construction, production support, and stakeholder deliverables, but priorities were not always expressed as shared, measurable outcomes.
 - T** I needed to help the group turn competing work into an aligned plan.
 - A** I facilitated goal-setting sessions, grouped work by business and platform outcome, clarified owners, dependencies, and completion criteria, and introduced a recurring cadence for checking progress and revising the plan.
 - R** Team members had a common view of priorities and tradeoffs, blockers surfaced earlier, and status conversations focused on outcomes rather than disconnected task lists.
- Implemented Jira and GitHub workflows to centralize task tracking and improve coordination across the analytics team.
 - S** Requests, implementation details, and review status were spread across conversations and tools, making priorities and ownership difficult to track.
 - T** I needed to create a visible path from a business request to reviewed production code.
 - A** I organized work into Jira epics and issues, connected tickets to GitHub branches and pull requests, introduced consistent issue and review expectations, and aligned workflow states with the team's delivery process.
 - R** The analytics team gained clearer ownership and status, reviewers could trace changes back to their requirements, and handoffs involved less searching and follow-up.
- Mentored junior team members in code review, testing, pipeline design, and delivery practices.
 - S** Junior team members were taking on increasingly complex SQL, dbt, and AWS work and needed support beyond one-time corrections.
 - T** I wanted to help them develop an independent method for designing, testing, and shipping reliable changes.
 - A** I paired on model and pipeline design, used reviews to explain grain, edge cases, and test strategy, demonstrated structured debugging, and gradually shifted implementation and review ownership back to them.
 - R** They became more confident diagnosing failures and preparing changes, required less step-by-step support, and contributed work that aligned more consistently with team delivery standards.

Jul. 2022 - Mar. 2023

Merative**Senior Data Scientist**

Technologies: Microsoft Azure, Azure Data Factory, Blob Storage, Azure Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps

Served as the quality and acceptance layer for a large scale healthcare platform migration from HDFS to Azure. Built automated reconciliation, regression, deidentification, and governance controls that brought the tested data to sign-off with no unresolved discrepancies.

- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.
 - S** The migration included demographic, encounter, and coded clinical data representing approximately 300 million patient lives, so a deidentification defect could create both privacy exposure and analytical breakage.
 - T** I needed to validate that protected fields were transformed correctly while required relationships and clinical utility remained intact.
 - A** I translated deidentification requirements into field-level tests, checked representative known values and edge cases, scanned for residual identifying patterns, verified consistency across related records, and traced every failure back to its transformation or configuration.
 - R** All in-scope deidentification checks passed before sign-off, providing evidence that the migrated data met the defined HIPAA-compliant handling requirements.
- Engineered a seven-layer Python and SQL migration-acceptance suite with zero unresolved discrepancies at sign-off across tested record counts, partitions, schemas, nulls, duplicates, field values, aggregates, and business rules.
 - S** Merative was moving a large healthcare data platform from HDFS to Azure, and successful job completion alone could not prove that the new platform preserved the source data's meaning.
 - T** I needed to create a repeatable acceptance framework that could support a defensible migration sign-off.
 - A** I built a seven-layer Python and SQL suite that compared source and target record counts, partitions, schemas, null and duplicate behavior, field-level values, aggregates, and business rules. I recorded expected tolerances, produced rerunnable discrepancy reports, and worked each unexplained variance through defect, retest, and closure.
 - R** The tested migration scope reached sign-off with zero unresolved discrepancies and a reproducible evidence trail for the decision.

- Enforced data-governance and security protocols supporting audit-risk reduction and HITRUST readiness.
 - S** Moving regulated healthcare data to Azure changed where data was stored, who could access it, and which controls had to be evidenced for internal review and HITRUST readiness.
 - T** I needed to incorporate governance and security checks into migration acceptance rather than treat them as a separate final review.
 - A** I verified that sensitive datasets followed approved storage and handling patterns, checked role-based access and least-privilege expectations, confirmed encryption and logging controls, reviewed retention and movement rules, and maintained evidence and exception records through remediation.
 - R** The migration closed control gaps before sign-off, reduced audit risk, and produced a more complete evidence set for the organization's HITRUST-readiness work.
- Collaborated with multidisciplinary engineering teams to refine architectural designs and preserve workflow continuity and data fidelity.
 - S** The migration crossed teams responsible for the HDFS source, Azure services, data engineering, security, and downstream healthcare analytics, and a local design improvement could disrupt another team's workflow.
 - T** I needed to use validation evidence to help refine the architecture without losing continuity or fidelity.
 - A** I joined design and defect reviews, mapped dependencies and source-to-target contracts, presented concrete reconciliation results and failing examples, clarified acceptable differences, and retested proposed changes across upstream and downstream boundaries.
 - R** Architectural issues were resolved before cutover, teams made decisions from shared evidence, and existing analytical workflows retained the data relationships and behavior they depended on.
- Validated end-to-end Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations during the transition from HDFS to Azure.
 - S** Individual transformations could pass while an end-to-end workflow still lost a partition, used the wrong storage path, or delivered incomplete data to Synapse.
 - T** I needed to validate the complete path from HDFS extraction through the Azure analytical target.
 - A** I ran controlled cohorts through Azure Data Factory, verified triggers, dependencies, parameters, retries, and failure behavior, reconciled Blob Storage paths and file or partition counts, and queried the resulting Synapse tables for completeness and transformation accuracy. I repeated the checks after configuration and code changes.
 - R** The team had evidence that each handoff worked as a connected system, not merely as independently successful components.
- Designed and implemented automated ETL testing and regression suites that detected schema drift and transformation errors early in the migration lifecycle.
 - S** Source schemas and migration transformations continued to change while the Azure platform was being built, so defects discovered only during final acceptance would be expensive to isolate.
 - T** I needed to move quality checks earlier in the delivery cycle.
 - A** I established versioned schema and data-profile baselines, automated checks for columns, types, keys, nulls, duplicates, distributions, and derived fields, and connected the regression suite to repeatable pipeline test runs. Material differences generated focused evidence for the responsible engineering team rather than a generic failed comparison.
 - R** Schema drift and transformation errors were detected closer to the change that caused them, reducing late-cycle investigation and protecting the final acceptance schedule.
- Documented test plans, platform-migration procedures, and onboarding guidance for future quality-assurance engineers.
 - S** The acceptance process accumulated detailed knowledge about source behavior, Azure configuration, expected differences, and recovery steps that could easily remain with the original migration team.
 - T** I needed to make that process repeatable for future quality-assurance engineers.
 - A** I documented test scope and prerequisites, source-to-target mappings, execution steps, expected results and tolerances, discrepancy triage, rerun and escalation procedures, and the evidence required for sign-off. I organized the material into onboarding guidance that followed the actual validation workflow.
 - R** A new engineer could understand the platform and execute the core checks from documented procedures rather than reconstructing the migration logic from code and conversations.

IBM

Jun. 2015 - Jul. 2022

Progressed through five increasingly senior roles over seven years, evolving from hands on healthcare data integration into technical leadership across platform engineering, data architecture, governance, analytics, and cloud migration validation. Repeatedly converted bespoke client problems into reusable tooling, governed platforms, validation frameworks, and team practices.

Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, T-SQL, MySQL, PostgreSQL, PL/SQL, MQL, XPath, XQuery, Apache Pig, Jaql, JSONPath, Docker, Jenkins, Tableau, HL7

- Served on tiger teams supporting the HDFS-to-Azure data-platform upgrade and participated on the data-governance team.
 - S** The HDFS-to-Azure upgrade affected shared clinical data, downstream applications, and governance controls across multiple teams.
 - T** I was asked to help resolve cross-cutting migration risks and preserve trusted data behavior.
 - A** On tiger teams, I traced source-to-target dependencies, analyzed validation failures, documented platform constraints, and brought findings into governance discussions about lineage, access, retention, and quality. I then retested changes against affected workflows.
 - R** The program could make architecture and governance decisions from concrete data evidence, and high-risk issues were surfaced before they reached migration sign-off.
- Played a central role in the REACH cross-team training initiative, teaching SQL fundamentals to nontechnical colleagues and ETL, architecture, data-science, and predictive-modeling practices to technical colleagues.
 - S** IBM colleagues needed different levels of data fluency, from basic SQL interpretation to practical ETL, architecture, data-science, and modeling skills.
 - T** I needed to make cross-team instruction useful to both nontechnical and technical audiences.
 - A** Through REACH, I separated the curriculum by audience, built examples around familiar healthcare-data problems, combined explanation with hands-on exercises, and adjusted later sessions from participant questions and observed gaps.
 - R** Colleagues gained a shared vocabulary and practical techniques they could apply in project discussions, analysis, and delivery rather than receiving one generic training course.
- Supported employee onboarding from 2016 to 2018, helping new colleagues learn team practices, systems, and technical workflows.
 - S** Rapid growth brought new colleagues into a complex healthcare data environment where platform conventions and delivery practices were largely learned through experience.
 - T** I needed to help new hires become useful contributors without relying on unstructured shadowing.
 - A** I explained the platform and team workflow, paired on representative integration tasks, reviewed early work, connected people with domain experts, and used recurring questions to improve onboarding material between 2016 and 2018.
 - R** New colleagues had a clearer path from orientation to independent delivery, and the team spent less time repeatedly reconstructing the same foundational knowledge.
- Participated in internal coding jams, hackathons, and cross-team delivery initiatives.
 - S** Teams working on different products often encountered similar ingestion, analytics, and workflow problems but had few opportunities to experiment together outside committed delivery.
 - T** I wanted to test ideas quickly and build connections across specialties.
 - A** I joined coding jams, hackathons, and cross-team initiatives, helped narrow broad ideas into demonstrable use cases, contributed data preparation and analytical code, and presented results and limitations to colleagues.
 - R** The events produced reusable technical approaches and working relationships that informed later project delivery, even when a prototype itself did not become a production product.

Senior Data Scientist

Nov. 2021 - Jul. 2022

Led healthcare data integration, migration validation, and analytics delivery across clinical, claims, operational, and financial sources, including an HDFS-to-Azure transition spanning more than 150 billion deidentified records.

- Served as technical lead across electronic health records, insurance claims, scheduling and financial data, social determinants of health, and other relevant sources.
 - S** Provider analytics questions increasingly crossed EHR, claims, scheduling, finance, and social-determinants data, none of which shared a simple common model.
 - T** I needed to lead the integration while preserving each source's meaning and limitations.
 - A** I defined contracts and analytical grains, established identity and time-window rules, documented lineage and caveats, coordinated implementation across owners, and required cross-source reconciliation.
 - R** Analysts received an integrated view they could use without hiding material differences in coverage, timing, or provenance.
- Developed migration-validation analysis spanning more than 150 billion deidentified health records, reconciling source and target schemas, partitions, values, aggregates, and business rules during the transition from legacy HDFS infrastructure to Azure.
 - S** Moving more than 150 billion deidentified health records from HDFS to Azure risked changing content or business behavior despite successful loads.
 - T** I needed migration evidence at a scale where manual sampling was inadequate.
 - A** I automated source-to-target comparisons for schemas, partitions, counts, field values, aggregates, and business rules; isolated differences by dataset and stage; and tracked unexplained variances through correction and rerun.
 - R** The team could distinguish expected platform differences from defects and make acceptance decisions using repeatable, documented evidence.

- Supported data and integration work for Pennsylvania's statewide P3N health-information exchange, encompassing master patient identity, provider directories, clinical-document discovery, patient consent, and real-time ADT encounter notifications.
 - S** Pennsylvania's P3N exchange had to connect patient identity, provider directories, consent, document discovery, and real-time ADT notifications across organizations.
 - T** I needed to support the connected workflow rather than treat each function as an isolated feed.
 - A** I mapped identifiers and message flows, validated representative patient and provider matches, checked consent and document-discovery behavior, traced ADT events end to end, and documented exceptions.
 - R** The exchange gained clearer evidence that workflows preserved identity, authorization, and event continuity across boundaries.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.
 - S** Prior-authorization teams lacked a consistent view of where requests waited and which stages created bottlenecks.
 - T** I needed to turn large workflow histories into reusable timing measures.
 - A** I built a Spark, Hive, and Impala framework that normalized status events, calculated stage and end-to-end lag, preserved historical snapshots, and exposed shared dimensions and measures to more than 50 reports.
 - R** Operations received daily bottleneck visibility while regulatory and financial users received consistent periodic snapshots from the same governed logic.
- Worked with senior leaders and colleagues to identify opportunities, define objectives, and formulate analytics strategies that produced measurable impact.
 - S** Senior leaders had multiple promising analytics ideas, but their value, feasibility, and measures of success were not always explicit.
 - T** I needed to turn broad opportunities into executable analytical work.
 - A** I facilitated discovery conversations, translated business concerns into testable questions, identified available data and constraints, defined outcome measures and decision points, and proposed phased strategies.
 - R** Stakeholders could compare opportunities on a common basis, and the team focused on analyses with clear objectives, owners, and a credible path to measurable impact.
- Served as a lead member of Provider Analytics, integrating disparate and centralized sources into data marts and supporting data mining, statistical-model development, and advanced analytics.
 - S** Provider Analytics depended on clinical, claims, and operational sources that used different identifiers, grains, and refresh schedules.
 - T** As a lead member, I needed to create dependable data marts for analysts and modelers.
 - A** I clarified analytical grains, mapped source identities to conformed entities, standardized measures and time windows, built reconciliation checks, and worked with analysts to validate that the marts supported mining and statistical-model use cases.
 - R** The team gained reusable, traceable datasets instead of rebuilding source logic for each analysis, shortening the path from question to result.
- Supported the controlled decommissioning of legacy Explorys capabilities by analyzing downstream dependencies and validating continuity across normalized clinical, claims, laboratory, billing, and operational datasets.
 - S** Retiring legacy Explorys capabilities risked breaking hidden dependencies across clinical, claims, laboratory, billing, and operational workflows.
 - T** I needed to identify what could be retired safely and what required a replacement path.
 - A** I inventoried consumers, traced lineage to legacy datasets, grouped dependencies by criticality, compared replacement outputs to current behavior, and supported staged validation before shutdown.
 - R** The program could decommission capabilities in a controlled sequence and expose unresolved dependencies before access disappeared.
- Supported Cleveland Clinic Foundation's accelerated SuperMart migration from MySQL to Greenplum by assessing dashboard dependencies and defining the changes required to preserve reporting continuity.
 - S** Cleveland Clinic Foundation needed to accelerate a SuperMart migration from MySQL to Greenplum while dashboards depended on database-specific tables and SQL behavior.
 - T** I needed to define the compatibility work required to preserve reporting.
 - A** I inventoried dashboard queries and source objects, identified differences in types, functions, joins, and performance, mapped dependencies to target changes, and prioritized validation by criticality.
 - R** The team received an actionable conversion plan and avoided finding essential dashboard dependencies only after cutover.

- Led delivery of a tailored analytics and reporting project using client administrative data.
 - S** A client had administrative data but lacked reporting tied to the operational decisions it needed to make.
 - T** I needed to lead a tailored delivery without imposing a generic dashboard.
 - A** I clarified the decisions and audience, profiled the fields, defined the reporting grain and measures, built a validated analytical dataset, and iterated on reports using examples and reconciliation totals.
 - R** The client received reporting aligned to its workflow, with documented metrics and a traceable path back to source data.
- Provided production and configuration support for Phytel population-health applications, troubleshooting data and operational issues affecting provider configuration, care-gap identification, patient outreach, and quality-reporting workflows.
 - S** Phytel issues could appear as missing care gaps, incorrect provider populations, failed outreach, or reporting discrepancies even when the root cause was configuration or data.
 - T** I needed to restore workflows and identify the right lasting fix.
 - A** I traced examples from provider configuration through patient and measure data, compared expected and actual states, reviewed recent changes, and coordinated corrections across support and engineering.
 - R** Operational issues were resolved with a documented cause, and recurring defects could be prevented through targeted configuration or data changes.
- Created and delivered Phytel configuration training covering provider activation, system changes, support procedures, and the clinical-data workflows underlying proactive population-health management.
 - S** Phytel administrators needed to understand how configuration changes affected care-gap, outreach, and population-health workflows.
 - T** I needed to teach both the procedure and its downstream data consequences.
 - A** I built scenario-based training, demonstrated activation and change procedures, traced examples into clinical workflows, covered common failure paths, and used exercises to confirm understanding.
 - R** Learners could perform routine configuration more confidently and recognize when an issue required data, application, or engineering support.
- Developed and delivered peer-led Azure instruction through the REACH initiative, translating cloud architecture, storage, data integration, security, and analytics concepts into practical guidance for technical colleagues.
 - S** Colleagues familiar with the legacy platform needed a practical way to understand Azure, not a catalog of services.
 - T** Through REACH, I needed to connect cloud concepts to work they already performed.
 - A** I designed instruction around architecture, storage, integration, security, and analytics, mapped legacy patterns to Azure equivalents, used migration exercises, and revised material from participant questions.
 - R** Colleagues gained a working mental model for evaluating Azure designs and could participate more effectively in migration discussions.
- Performed postmigration validation for an enterprise Jira transition, reconciling projects, issues, workflows, custom fields, permissions, histories, and relationships and documenting discrepancies for remediation before cutover.
 - S** A Jira migration could appear complete while losing workflow states, fields, permissions, history, or links that teams relied on.
 - T** I needed to validate functional equivalence before cutover.
 - A** I built source-to-target inventories and counts for projects, issues, workflows, fields, permissions, histories, and relationships; tested complex cases; categorized findings by severity; and documented reproducible defects for remediation and retest.
 - R** The team had acceptance evidence and a clear defect list, reducing the risk of missing access or broken work history after migration.

Senior Data Architect

Dec. 2020 - Nov. 2021

Directed data-onboarding architecture and governance for large-scale clinical platforms, combining metadata-driven integration, deidentification, lineage, validation, and applied machine learning.

- Served as a subject-matter expert to project data architects and delivery, analytics, and integration teams; directed configuration of the data-onboarding platform and supporting infrastructure.
 - S** Project architects and delivery, analytics, and integration teams made interconnected configuration decisions across the onboarding platform.
 - T** I needed to provide a consistent technical point of view and direct configurations toward reusable patterns.
 - A** I reviewed source designs, clarified grains and mappings, identified infrastructure dependencies, resolved cross-team questions, and required validation evidence before configurations advanced.
 - R** Teams made faster decisions with fewer contradictory assumptions, and onboarding configurations aligned more consistently with platform standards.

- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.
 - S** Approximately 200 integrations used static source mappings that were expensive to update and difficult to audit consistently.
 - T** I needed to separate integration instructions from execution logic.
 - A** I designed metadata-driven instruction sets, standardized EDI batch schemas, built reusable transformation behavior around that metadata, and added Python/Jupyter lineage tools and validation gates from source through output.
 - R** New and changed mappings could be delivered through governed configuration, reducing duplicated code and making lineage and validation more consistent across integrations.
- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.
 - S** Four EPM applications, more than 30 risk models, and hundreds of measures depended on over 100 billion clinical records from heterogeneous sources.
 - T** I needed to design pipelines that could scale without losing lineage or consistent analytical meaning.
 - A** I defined layered Spark and Hadoop processing, partition and schema strategies, metadata and identity rules, and validation gates for structured and semi-structured inputs; I then tuned heavy stages using observed execution behavior.
 - R** The platform supplied reusable, governed datasets at the scale required by applications, models, measures, and registries.
- Led a HIPAA patient-deidentification project drawing from more than 100 data sources for content analytics and measures.
 - S** Content analytics and measures required data from more than 100 sources, each with different identifiers and potential protected-health-information exposure.
 - T** I needed to lead a common deidentification approach that protected privacy while preserving analytical relationships.
 - A** I inventoried sensitive fields, translated policy into source-specific transformations, standardized pseudonymous identities, tested for residual identifiers and broken linkages, and coordinated exceptions with governance and engineering.
 - R** The project delivered a consistent deidentified foundation for downstream analytics with documented controls and traceability.
- Led a cross-functional team as the clinical data analyst for the 2020/2021 measure-updates project.
 - S** Annual 2020/2021 measure updates changed clinical definitions, value sets, and data requirements across products and teams.
 - T** As the clinical data analyst, I needed to coordinate the impact from specification through validated release.
 - A** I decomposed measure changes, mapped them to source fields and existing logic, organized work across engineering and clinical stakeholders, reviewed edge cases, and reconciled updated outputs against expected populations.
 - R** The cross-functional team delivered the update with a shared interpretation of the specifications and traceable validation for changed measures.
- Served as a senior member of Data Onboarding & Governance, integrating healthcare data, enriching metadata, and configuring data models, analytics, reports, and visualizations.
 - S** New healthcare datasets had to become usable across models, analytics, reports, and visualizations while meeting governance expectations.
 - T** As a senior Data Onboarding and Governance member, I needed to connect technical ingestion with downstream meaning.
 - A** I profiled sources, mapped entities and metadata, configured data models and analytical outputs, documented lineage and constraints, and validated results with delivery and analytics teams.
 - R** Onboarded data arrived with a clearer contract and provenance, giving downstream teams a more reliable foundation for reporting and analysis.
- Served in an advisory capacity for development of a new data platform.
 - S** A new data platform was being designed while teams still depended on legacy integration patterns and healthcare-specific workflows.
 - T** In an advisory role, I needed to bring practical delivery constraints into architecture decisions without owning the entire build.
 - A** I reviewed proposals, translated legacy lessons into requirements, identified migration and governance risks, challenged assumptions with representative workloads, and recommended validation and lineage capabilities.
 - R** The platform team could evaluate design choices against real operating needs and address important compatibility and control gaps earlier.

- Developed a topic-modeling project using nearly six million semi-structured provider notes.
 - S** Nearly six million provider notes contained useful themes that were difficult to discover through manual review or fixed keyword searches.
 - T** I needed to test whether topic modeling could organize the corpus for analyst exploration.
 - A** I cleaned and normalized note text, created a modeling sample and representation, compared topic counts and stability, reviewed representative terms and documents, and worked with domain users to label useful themes and reject incoherent ones.
 - R** The project produced an exploratory map of recurring clinical themes and demonstrated where unsupervised methods could assist, but not replace, expert review.
- Developed a surgical-scheduling optimizer and CNN/RNN case-card prediction proof of concept that was being evaluated as a potential new offering.
 - S** Surgical teams faced schedule inefficiency and uncertainty about the supplies or case-card content needed for upcoming procedures.
 - T** I needed to test whether optimization and sequence models could improve planning enough to justify a product offering.
 - A** I modeled scheduling constraints and objectives, prepared historical procedure and case-card sequences, built an optimizer plus CNN/RNN prediction prototypes, and evaluated feasibility, accuracy, and operational tradeoffs with stakeholders.
 - R** The proof of concept gave product leaders concrete performance evidence and implementation risks for deciding whether to invest in a new offering.
- Created an exploratory machine-learning COVID-19 prototype to surface hospitalization-risk signals for retrospective analyst review.
 - S** Early in the COVID-19 response, analysts needed a rapid way to explore which historical patient characteristics were associated with hospitalization.
 - T** I needed to create an exploratory prototype without presenting it as a clinical decision system.
 - A** I defined a retrospective cohort and outcome, prepared candidate features, trained baseline classification models, examined performance and important signals, and documented limitations such as bias, missingness, and temporal instability.
 - R** Analysts gained a structured set of risk signals for further review and a clear boundary between exploratory evidence and a deployable clinical model.
- Closed project-documentation gaps by organizing material from more than 100 past projects.
 - S** Useful decisions, mappings, and troubleshooting knowledge from more than 100 projects were scattered across inconsistent locations and formats.
 - T** I needed to preserve what was reusable and make it discoverable.
 - A** I inventoried available material, grouped it by workflow and subject, removed duplication, identified missing context, established naming and navigation conventions, and highlighted patterns and lessons that applied across clients.
 - R** Teams gained a more usable project knowledge base, reduced repeated investigation, and had better starting material for onboarding and future delivery.

Data Architect

Nov. 2017 - Dec. 2020

Led healthcare-data onboarding and validation across complex multi-EHR integrations, strengthening data-sharing controls, patient-matching quality, and repeatable client delivery.

- Led a multi-organization healthcare integration spanning 70+ EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules and coordinating delivery with engineering teams.
 - S** A multi-organization integration combined more than 70 EHR sources, but data-sharing rules permitted patient data only within appropriate provider relationships.
 - T** I needed to enforce that scope without omitting eligible patients.
 - A** I mapped provider and patient attribution rules, implemented provider-scoped filtering early in the pipeline, built leakage and completeness tests, reviewed edge cases with stakeholders, and coordinated changes across engineering teams.
 - R** The integration applied a consistent data-sharing boundary across heterogeneous sources while preserving traceable evidence for included and excluded records.
- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.
 - S** Client integrations repeatedly reached review with preventable mapping and validation defects, creating Jira rework late in delivery.
 - T** I needed to move consistent checks ahead of client review across approximately 100 integrations.
 - A** I built reusable SQL validation patterns for counts, keys, required fields, mappings, and business rules, defined review gates and evidence, and fed recurring defects back into templates and guidance.
 - R** Jira- tracked validation rework fell 25% and mapping rework fell 45%, giving teams faster, more predictable client review cycles.

- Built the data-onboarding platform and associated infrastructure, remediated UAT data issues, and reported on data health during onboarding.
 - S** New clients needed a repeatable route from raw source delivery to usable platform data, while UAT defects and unclear data health slowed acceptance.
 - T** I needed to establish the onboarding foundation and make readiness visible.
 - A** I configured platform and infrastructure components, defined source-to-target stages and checks, triaged UAT examples to the responsible mapping or transformation, and summarized completeness, quality, and open issues in data-health reporting.
 - R** Stakeholders could see what was ready, what was blocked, and why, and the team had a more repeatable process for moving integrations through UAT.
- Earned AI certification and peer-reviewed patient-matching algorithm tuning across the project portfolio.
 - S** Patient-matching changes could reduce duplicates but also create harmful false matches across many client projects.
 - T** I needed to apply stronger AI knowledge and an independent review discipline to tuning decisions.
 - A** After completing AI certification, I reviewed proposed features, thresholds, training or test cohorts, and error examples; compared precision and recall tradeoffs; and required project teams to document why a change was appropriate for their data.
 - R** Tuning decisions received a consistent peer review and clearer evidence before being applied across the portfolio.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.34%.
 - S** SaaS client delivery included recurring effort that limited the team's capacity to provide additional billable value.
 - T** I needed to improve the economics of each client without reducing service quality.
 - A** I reviewed where onboarding and support effort was being consumed, standardized repeatable validation and configuration work, and helped redirect recovered capacity toward higher-value client services and timely delivery.
 - R** Revenue per SaaS client improved by at least 1.34%, and the contribution was recognized through a GDP bonus.
- Received the Lab Services Award for coordinating internal project teams to deliver client needs on time.
 - S** A client delivery depended on several internal teams with separate priorities, creating schedule and handoff risk.
 - T** I needed to coordinate the work so the client received the required outcome on time.
 - A** I translated needs into shared milestones, clarified owners and dependencies, surfaced blockers early, connected technical decisions to client impact, and maintained a focused cadence through delivery and validation.
 - R** The teams delivered the client need on schedule, and IBM recognized the coordination and client impact with the Lab Services Award.
- Curated 25 pages of AI-workflow documentation and mentored new hires in the AI-tuning process.
 - S** Patient-matching tuning depended on specialized knowledge that new hires could not safely learn from parameter names alone.
 - T** I needed to make the workflow teachable without oversimplifying its risk.
 - A** I curated 25 pages covering data preparation, tuning steps, evaluation measures, error review, approval points, and common failure modes, then paired with new hires on representative cases and reviewed their reasoning.
 - R** New team members gained a structured path into AI tuning and could contribute with better consistency and awareness of matching tradeoffs.
- Spearheaded a new-hire mentoring program that standardized technical ramp-up and prepared colleagues to contribute to billable client delivery.
 - S** New hires entered with varied healthcare and platform experience, and informal mentoring produced uneven readiness for client work.
 - T** I needed to standardize technical ramp-up while preserving individualized support.
 - A** I organized a mentoring program with role-based learning objectives, guided exercises, shadowing, code and design reviews, and readiness checkpoints tied to representative delivery tasks.
 - R** Managers had a clearer signal of when colleagues were ready, and new hires reached billable work with a more consistent foundation and support network.

Software Developer, Data Platform

Sep. 2016 - Nov. 2017

Developed healthcare data-platform tooling and training programs that automated HL7 ingestion, reduced vendor costs, and accelerated knowledge transfer.

- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating \$30,000 in annual vendor costs.
 - S** Approximately 30 internal users depended on a Mirth-based process to parse HL7 v2 and v3 data, creating licensing cost and limiting control over changes.
 - T** I needed to replace the workflow without disrupting existing consumers.
 - A** I documented supported message structures, built a Pig pipeline to parse and normalize segments and fields, compared outputs against the Mirth results, added malformed-message handling, and supported a controlled transition with user guidance.
 - R** Internal users retained the required HL7 data while IBM eliminated approximately \$30,000 in annual vendor costs.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.17%.
 - S** Repeated integration and support effort constrained how much value the team could deliver per SaaS client.
 - T** I needed to reduce avoidable effort and expand capacity for billable work.
 - A** I automated a recurring technical workflow, strengthened documentation and training, and standardized delivery steps so specialists spent less time on manual setup and repeated troubleshooting.
 - R** The recovered capacity contributed to at least a 1.17% improvement in revenue per SaaS client and was recognized through a GDP bonus.
- Received the Eminence and Excellence Award in 2017 for an analytics project.
 - S** A project team had a large operational dataset but lacked a concise analytical view of the patterns driving a client problem.
 - T** I needed to turn the data into a defensible analysis that stakeholders could act on.
 - A** I clarified the business question, prepared and validated the analytical dataset, tested alternative explanations, presented the important patterns with their limits, and packaged the method so the analysis could be repeated.
 - R** The work informed the project team's decisions and was recognized with IBM's 2017 Eminence and Excellence Award.
- Rewrote and reorganized project documentation to streamline knowledge transfer.
 - S** Project documentation had accumulated around components and historical decisions rather than the workflow a new contributor needed to follow.
 - T** I needed to make knowledge transfer faster and less dependent on the original developers.
 - A** I inventoried existing material, removed stale or duplicate guidance, reorganized it around setup, architecture, operating tasks, troubleshooting, and escalation, and validated the instructions by following them from a new-user perspective.
 - R** Contributors could find the right procedure more quickly, and handoffs required less repeated explanation from experienced team members.
- Redeveloped the employee education program to make new-hire training more efficient and repeatable.
 - S** New-hire education varied by mentor and project assignment, so important platform concepts were taught inconsistently.
 - T** I needed to turn training into a repeatable program that still prepared people for real delivery.
 - A** I organized role-based modules, sequenced foundational and advanced topics, created hands-on exercises and reference material, added checkpoints and a representative capstone task, and incorporated learner and mentor feedback.
 - R** New hires received a clearer progression, mentors could reuse the same structure, and the team spent less time rebuilding training for each cohort.

Data Scientist

Nov. 2015 - Sep. 2016

Directed client data-integration delivery and built reusable extraction and QA practices that reduced implementation effort and standardized onboarding across dozens of healthcare projects.

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
 - S** Thirty-six client integrations were being delivered with 12 remote contractors, making quality, communication, and issue ownership inconsistent.
 - T** I needed to coordinate parallel delivery without becoming the only person who could resolve every problem.
 - A** I clarified work packages and owners, established code-review and QA expectations, introduced a common issue-resolution path, reviewed high-risk mappings, and used recurring defects to improve standards.
 - R** Contractors could deliver against shared expectations, issues reached the right owner faster, and integrations progressed with more consistent technical quality.

- Served as principal integration specialist on more than 60 projects for 20 clients across the broader 2015–2018 data-ingestion period.
 - S** More than 60 projects for 20 clients brought different systems, healthcare workflows, and delivery schedules, but many integration problems repeated.
 - T** As principal specialist, I needed to solve the immediate client issue while building reusable expertise.
 - A** I led discovery, profiled source data, defined mappings and acceptance checks, diagnosed ingestion failures, and turned recurring patterns into tools, documentation, and review guidance.
 - R** Teams could apply proven approaches across clients, reducing reinvention while retaining the source-specific analysis each integration required.
- Built a Python, Ruby, and SQL extraction tool that eliminated unnecessary story points and reduced average engineering hours per client integration.
 - S** Engineers spent repeated effort inspecting source systems and assembling the same metadata and extracts before integration work could begin.
 - T** I needed to remove avoidable manual steps from each client onboarding.
 - A** I built a Python, Ruby, and SQL tool that connected to supported sources, extracted required structures and sample information, normalized the output, and produced artifacts developers could use for mapping and estimation.
 - R** The tool eliminated unnecessary story points and lowered average engineering hours per integration while making discovery output more consistent.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.
 - S** Integration developers repeatedly asked the same setup, mapping, and troubleshooting questions because answers were scattered in tickets and personal notes.
 - T** I needed to make operational knowledge searchable and reusable.
 - A** I identified recurring questions, wrote articles with context, prerequisites, steps, examples, expected results, and failure paths, linked related topics, and updated guidance when platform behavior changed.
 - R** I produced more than 125 articles—about 30% of the team's published content—giving the organization a consistent reference for development and incident resolution.
- Eliminated an average of three weeks from employee onboarding.
 - S** New integration specialists spent too long locating documentation, gaining system access, and learning the sequence from source discovery to delivery.
 - T** I needed to shorten ramp-up without skipping essential practice.
 - A** I reorganized onboarding around a defined learning path, consolidated reference material, introduced guided exercises and a representative integration, paired new hires with reviewers, and used checkpoints to target remaining gaps.
 - R** Employees reached productive project work an average of three weeks sooner, while managers retained a structured way to assess readiness.

Explorys, an IBM Company

Jun. 2015 - Nov. 2015

Data Scientist

Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins

Joined Explorys during a period of rapid client expansion and its transition into IBM Watson Health. Helped scale healthcare data onboarding across more than 10 client warehouses by leading source discovery and integration setup, diagnosing Hadoop and Oracle ingestion failures, and introducing scheduling, recovery, and blackout controls for production data feeds. Contract via Ranstad USA.

- Identified and extracted data from heterogeneous client systems; led technical requirements gathering, discovery, documentation, and implementation; and coordinated acquisition, installation, and configuration of VPN hardware supporting client integrations.
 - S** Each new healthcare client exposed different databases, schemas, network constraints, and operational contacts, so onboarding could not begin with a standard extract alone.
 - T** I needed to carry integrations from technical discovery through a secure, validated initial data flow.
 - A** I led source and requirements sessions, inventoried tables and refresh behavior, wrote extraction and mapping documentation, implemented and validated the data pull, and coordinated procurement, installation, configuration, and connectivity testing for required VPN hardware.
 - R** Client teams and Explorys had a shared integration contract, secure connectivity, and a repeatable path from heterogeneous source data to the onboarding platform.

- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.
 - S** Rapid client growth left more than 10 data warehouses dependent on ingestion jobs that could fail at the Oracle source, during transfer, or in Hadoop, with inconsistent recovery steps.
 - T** I needed to make failures diagnosable and restore feeds without rerunning more work than necessary.
 - A** I traced failed loads through source queries, transfer logs, HDFS landing data, and downstream jobs; documented recurring causes; and implemented Jenkins scheduling, dependency, restart, and notification controls around the ingestion workflow.
 - R** Production support gained a consistent way to identify the failed stage and recover client feeds, reducing reliance on ad hoc manual restarts.
- Developed a cron-based control for defining blackout periods around client data pulls.
 - S** Scheduled data pulls could overlap client maintenance, backups, or other restricted periods, creating avoidable source contention and failed loads.
 - T** I needed a lightweight way to respect client-specific blackout windows without manually disabling and re-enabling each integration.
 - A** I built a cron-based control that read configured blackout periods before starting a pull, skipped or deferred ineligible runs, logged the decision, and resumed the normal schedule afterward. I tested boundary times and representative client schedules.
 - R** The platform could honor operating constraints consistently, reducing preventable collisions while preserving an auditable schedule for production support.

AmTrust Financial Services

Mar. 2014 - Jun. 2015

Software Engineer

Technologies: VB.NET, ASP.NET, C#, Web APIs, CSS, JavaScript, HTML5, SQL, PL/SQL, LINQ, R

Brought in to develop agent facing commercial fire insurance quoting and policy generation software. Grew into the principal developer for a new product offering, translating ISO rating rules into more than 65 web forms, APIs, and SQL services while strengthening testing, performance, architecture, and coordination between business analysts and developers.

- Implemented a new line of business in VB.NET and Microsoft SQL Server, served as principal developer for a new product offering, and contributed to two major software-architecture initiatives in 2015.
 - S** AmTrust wanted to introduce a new commercial-insurance offering whose rating rules, forms, and policy workflow did not fit the existing product unchanged.
 - T** As principal developer, I needed to turn the product concept into a maintainable VB.NET and SQL Server implementation.
 - A** I decomposed the line of business into shared and product-specific rules, designed the data and service changes, coordinated form and API delivery, reviewed implementation decisions, and carried lessons into two broader architecture initiatives.
 - R** The new offering gained a working technical foundation while its design contributed to more reusable platform architecture in 2015.
- Developed more than 65 ISO rating-based web forms and analyzed page timings in R to identify optimization targets and improve application performance.
 - S** A new commercial product required more than 65 web forms derived from ISO rating rules, and the resulting pages varied in complexity and response time.
 - T** I needed to deliver the rule-driven forms while identifying where users experienced avoidable delay.
 - A** I translated rating requirements into reusable form and validation patterns, implemented the supporting services and data calls, captured page-timing observations, and used R to compare distributions and locate slow outliers before profiling their rendering or query paths.
 - R** The product received broad ISO-based form coverage and a data-driven set of performance improvements rather than ad hoc optimization.
- Supported the technical integration of acquired businesses into the commercial policy platform.
 - S** Acquired businesses brought policy products, data, users, and operating workflows that differed from AmTrust's commercial platform.
 - T** I needed to help incorporate those businesses without interrupting active policy work.
 - A** I inventoried source applications and dependencies, mapped products and policy data to platform concepts, identified functional and technical gaps, supported interface and migration changes, and validated representative transactions through a phased integration plan.
 - R** The acquired operations gained a defined route onto the commercial-policy platform while material differences and continuity risks were addressed before broader rollout.

- Served as the working liaison between business analysts and developers, translating requirements, reporting progress, and helping prioritize delivery needs.
 - S** Business analysts understood insurance rules while developers understood the application, but ambiguous translation between them created scope and acceptance risk.
 - T** I needed to maintain a shared understanding throughout delivery.
 - A** I walked analysts through examples and edge cases, translated decisions into technical behavior and acceptance criteria, explained constraints to both groups, reported progress and blockers, and helped sequence work by customer and implementation impact.
 - R** Questions were resolved earlier, developers worked from clearer requirements, and stakeholders had a more reliable view of what would be delivered and when.
- Evaluated the advantages, risks, and implementation costs of requested features and recommended maintainable solutions grounded in object-oriented design principles.
 - S** Feature requests often arrived with an expected interface change but hidden consequences for rating logic, data, integrations, and future maintenance.
 - T** I needed to help stakeholders choose solutions based on total impact, not only initial coding effort.
 - A** I clarified the business outcome, traced affected components, estimated implementation and support costs, compared alternatives and failure risks, and recommended designs that used clear responsibilities, encapsulation, and reusable abstractions.
 - R** Product decisions had an explicit technical rationale, and the team avoided short-term changes that would have created disproportionate complexity later.
- Designed, developed, tested, and maintained full-stack commercial-insurance applications spanning VB.NET and ASP.NET interfaces, multi-tier components, Web APIs, and SQL data services.
 - S** Commercial-insurance quoting and policy generation crossed web forms, application services, rating logic, APIs, and database procedures, so a change in one tier could create defects elsewhere.
 - T** I needed to deliver features end to end while preserving consistent behavior across the application.
 - A** I traced requirements through the VB.NET and ASP.NET interfaces, multi-tier components, Web APIs, and SQL services; implemented the coordinated changes; added unit and integration coverage; and validated representative quote and policy scenarios before release.
 - R** The team could deliver commercial-insurance capabilities through a controlled full-stack process with clearer traceability and fewer handoff gaps.
- Created a unit- and integration-test suite for multi-tier MVC components, improving regression coverage and establishing stronger team testing practices.
 - S** Changes to multi-tier MVC components were difficult to verify without repeatedly exercising entire quote and policy workflows by hand.
 - T** I needed to create a safety net that covered both isolated logic and important tier interactions.
 - A** I separated testable responsibilities, wrote unit tests for rating and application behavior, built integration tests across controllers, services, and data access, created representative fixtures, and used failures to refine team review expectations.
 - R** Regression coverage improved, developers received faster feedback on changes, and automated testing became a stronger part of normal delivery practice.
- Coordinated developer schedules and troubleshoot production problems across multiple application environments.
 - S** Multiple application environments and concurrent releases created coordination risk, while production issues could involve code, configuration, data, or deployment differences.
 - T** I needed to keep planned work moving while restoring service safely when incidents occurred.
 - A** I coordinated developer assignments and dependencies, compared behavior across environments, reviewed logs and database state, reproduced failures with representative cases, organized the appropriate fix and validation, and documented follow-up actions.
 - R** Production problems reached the right owner faster, releases had clearer sequencing, and fixes were validated across environments before closure.
- Refactored and maintained the existing VB.NET insurance-rating codebase to improve readability, performance, and long-term maintainability.
 - S** The existing VB.NET rating code had accumulated intertwined rules, duplicated logic, and performance-sensitive paths that made changes risky.
 - T** I needed to improve maintainability without changing valid insurance outcomes.
 - A** I identified duplication and complex dependencies, separated calculation, presentation, and data responsibilities, introduced clearer interfaces and naming, expanded regression tests around representative ratings, and profiled the most expensive paths before optimizing them.
 - R** The code became easier to review and extend, high-use paths performed more efficiently, and future rating changes could be made with a stronger regression safety net.

PPG Industries

Apr. 2012 - Aug. 2013

Color Scientist*Technologies: VB.NET, C#, SQL, C++, FORTRAN, JMP, spectrophotometry, numerical methods*

Joined PPG as a color scientist and software specialist supporting the modernization of legacy automotive color-matching technology. Applied spectroscopy, radiative-transfer physics, statistical modeling, and numerical methods to improve matching and measurement consistency, then became a global liaison helping translate that work into software, standardized workflows, training, and operational adoption.

- Served as technical voice for color data-management software, supporting its global development, launch, training, adoption, and application.
 - S** A global color data-management product had to serve laboratories, operations, and customer-facing teams with different terminology and workflows.
 - T** I needed to represent the underlying color science throughout product development and help users apply the software correctly after launch.
 - A** I translated scientific requirements into software behavior, reviewed prototypes and test cases, resolved questions from regional teams, developed technical training and examples, and incorporated launch feedback into guidance and enhancements.
 - R** The product launched with a more consistent scientific model and users had a technical resource for training, adoption, and difficult application questions.
- Served as the global liaison among Color, Technical, Operations, and IT teams, translating color-science requirements into consistent systems, workflows, and communications.
 - S** Regional Color, Technical, Operations, and IT groups used different vocabulary, priorities, and local practices, creating inconsistent interpretations of the same color-science requirement.
 - T** I needed to establish a shared understanding that could be implemented globally without ignoring local constraints.
 - A** I facilitated cross-functional discussions, documented the scientific intent and decision rules, mapped regional workflows and system impacts, resolved conflicting assumptions, and circulated examples and decisions in a common format.
 - R** Teams worked from more consistent requirements and could coordinate system changes, operating practices, and communications with fewer regional surprises.
- Championed a more science-centered organizational approach to color by introducing analytical methods, shared standards, and evidence-based decision-making.
 - S** Color decisions were sometimes driven by local custom or individual experience, making it difficult to compare results and scale improvements across teams.
 - T** I wanted to strengthen the role of reproducible evidence without discounting the expertise of experienced practitioners.
 - A** I introduced common measurement and analysis methods, demonstrated them on active business questions, encouraged teams to document assumptions and acceptance criteria, and used shared standards to connect expert judgment with quantitative evidence.
 - R** Teams gained a more consistent language for evaluating color performance and a stronger basis for repeating, reviewing, and communicating technical decisions.
- Developed statistical models from established theory, experimental data, and novel approaches to improve color measurement, matching, and process decisions.
 - S** Established color theory explained much of the measurement process, but real production data also contained instrument, material, and process variation that theory alone did not capture.
 - T** I needed to build models that remained scientifically grounded while being useful for operational decisions.
 - A** I combined theoretical predictors with designed experiments and historical observations, explored additional features and interactions, validated performance on held-out or repeated measurements, and translated model limitations into decision thresholds.
 - R** The models provided more actionable estimates for measurement quality, match selection, and process control while making uncertainty explicit.
- Evaluated color processes, instrumentation, and data flows to identify technical and operational risks and develop practical mitigation strategies.
 - S** Color results could drift because of instrument condition, sample preparation, environmental effects, data transformations, or inconsistent operating steps.
 - T** I needed to identify where those risks entered the workflow and recommend controls that operations could realistically sustain.
 - A** I mapped the process from sample preparation through measurement, data transfer, and decision-making; reviewed failure modes and available evidence; ranked risks by impact and detectability; and proposed calibration, validation, monitoring, and escalation controls.
 - R** Stakeholders gained a practical risk view of the end-to-end color process and a prioritized set of mitigations rather than isolated equipment fixes.

- Managed a portfolio of color-related intellectual property and supported the translation of research into operational tools and practices.
 - S** Color research generated algorithms, methods, and experimental findings whose value could be lost if ownership, novelty, and practical application were not tracked.
 - T** I needed to help manage that portfolio while moving useful ideas from research into operations.
 - A** I cataloged candidate inventions and supporting evidence, coordinated technical review with research and IP stakeholders, distinguished protectable concepts from implementation know-how, and helped translate selected methods into software requirements, procedures, and training.
 - R** PPG had clearer visibility into its color-related intellectual assets and a more direct path from promising research to repeatable operational use.
- Led cross-functional color-science and marketing initiatives from concept through on-time, in-full delivery for the Automotive Refinish business.
 - S** Automotive Refinish initiatives often combined technical color claims, customer-facing materials, software behavior, and launch deadlines owned by different functions.
 - T** I needed to keep the scientific content accurate while moving the complete initiative to an agreed delivery date.
 - A** I defined the technical scope with color scientists and marketing partners, converted the work into milestones and owners, reviewed claims and demonstrations against the data, surfaced dependencies early, and coordinated final readiness checks.
 - R** The initiative was delivered on time and in full with aligned technical evidence, customer messaging, and operational support.
- Improved color-matching classification accuracy using numerical integration methods.
 - S** Automotive-refinish color matching depended on classifying candidate formulas from spectral and colorimetric data, but the existing calculation lost useful information when it approximated continuous response curves.
 - T** I needed to improve classification accuracy without making the production workflow impractically slow.
 - A** I reviewed the mathematical assumptions, implemented numerical integration over the measured curves, compared alternative integration schemes on representative panels, and validated the classifications against known matches and expert review.
 - R** The revised method separated close candidates more reliably and gave the matching team a stronger quantitative basis for selecting formulas.
- Built cross-instrument correlation software in VB.NET and JMP to improve measurement consistency and support statistical risk analysis.
 - S** Spectrophotometers at different laboratories and customer locations could report small but operationally important differences for the same color sample.
 - T** I needed to quantify those differences and make measurements more comparable across instruments.
 - A** I built a VB.NET workflow to ingest and align instrument readings, used JMP to model bias and variation by device and color region, generated correlation adjustments, and summarized residual risk for technical stakeholders.
 - R** Teams could identify out-of-family instruments, apply a consistent correlation approach, and make color decisions with a clearer understanding of measurement uncertainty.
- Cowrote color workflows and best practices for internal operations and external customers, supported the change-management efforts required for adoption.
 - S** New color tools and analytical methods would not improve results if internal teams and customers continued to follow inconsistent measurement and matching practices.
 - T** I needed to turn technical knowledge into usable workflows and support the behavior change required for adoption.
 - A** I cowrote step-by-step practices with operational experts, added decision points and troubleshooting examples, tested the guidance with representative users, supported training and rollout communications, and refined the material from field feedback.
 - R** Users received a clearer, repeatable way to apply the tools, and adoption discussions focused on observable practices rather than abstract scientific concepts.
- Informed marketing strategy and communications by connecting customer needs with technical capabilities and broad operational insight.
 - S** Marketing needed to describe color capabilities in terms that mattered to refinish customers without overstating what the technology or operating network could deliver.
 - T** I needed to connect customer problems, scientific capabilities, and operational realities into credible positioning.
 - A** I gathered questions from customer-facing and regional teams, mapped them to product and process capabilities, reviewed proposed claims against technical evidence, and supplied examples that explained the practical value and boundaries of the solution.
 - R** Marketing communications were better aligned with customer needs and technical reality, giving commercial teams a more credible story and clearer guidance on where the offering fit.

- Wrote a Visual C++ and VB.NET DLL wrapper for a loaned spectroscopic instrument and developed formulation methods for an automated color-matching facility.
 - S** PPG was evaluating a loaned spectroscopic instrument for an automated color-matching facility, but its native interface did not integrate with the existing VB.NET formulation environment.
 - T** I needed to establish reliable software control of the instrument and turn its measurements into usable formulation inputs.
 - A** I studied the vendor interface, wrote a Visual C++ DLL wrapper, exposed a stable VB.NET contract, handled connection and error states, and developed and tested formulation methods using controlled samples.
 - R** The team could exercise the instrument through the planned automation stack and evaluate an end-to-end measurement-to-formulation workflow before making a larger equipment commitment.
- Debugged a VB.NET and FORTRAN optimization routine for a numerical radiative-transfer solution.
 - S** A radiative-transfer optimizer combined a VB.NET application with a legacy FORTRAN numerical routine, and some inputs produced unstable or nonconvergent formulation results.
 - T** I needed to determine whether the problem came from interlanguage data exchange, numerical assumptions, or the optimization logic itself.
 - A** I traced representative cases across the VB.NET/FORTRAN boundary, verified array dimensions and parameter units, instrumented intermediate values, reproduced edge conditions, and corrected the faulty convergence and input-handling behavior.
 - R** The solver produced stable results across a broader operating range and became easier for the team to diagnose when unusual samples were encountered.

Selected Independent Projects

- iphone-cam-lite** 2025 - 2026
Native Swift/SwiftUI macOS menu bar application that selects an iPhone Continuity Camera and microphone in Zoom or Microsoft Teams, then reads back the application state to verify that the requested devices were actually selected. Features modular application adapters, permission-aware onboarding, system diagnostics, configurable device and accessibility labels, persistent settings, actionable error reporting, hermetic Swift tests, continuous integration, and scripted application packaging.
- farm-invoice-parser** 2025
Ruby command line prototype that converts handwritten or printed farm invoices into structured transaction records. The pipeline extracts text from invoice images with Tesseract OCR, sends the result to configurable language-model providers through RubyLLM, parses the response as JSON, and validates it against a versioned invoice schema. It also provides an OCR-only diagnostic mode and includes RSpec, RuboCop, and GitHub Actions scaffolding.
- cfd_snow_fence** 2025
Python applied modeling experiment for evaluating potential snow fence locations using measured wind data. The project maps wind vectors onto a two-dimensional spatial grid, estimates pressure fields through a finite difference Navier–Stokes/Poisson calculation, and aggregates the resulting simulations. It then trains an XGBoost classifier using position, wind speed, and pressure to identify low wind regions as a proxy for potential snow drift accumulation.
- claims_case_study** 2023
End to end healthcare workforce planning case study analyzing Cleveland area clinician, facility, and claims data for 2021–2022. The project evaluates source data quality, converts the original workbook into schema preserving Parquet datasets, standardizes the analytical data, and performs exploratory and time series analysis. ARIMA forecasts with error diagnostics project 2023 demand for Internal Medicine, Urology, and Anesthesiology, culminating in a written technical report and stakeholder oriented presentation.
- HL7-ingestion** 2023
Java 21 Spring Boot/Jersey service that accepts HL7 v2 messages as multipart file uploads or plain text and converts them into JSON encoded FHIR R4 collection Bundles using the LinuxForHealth converter. The project packages the service with Docker and a Helm chart for Kubernetes deployment, includes GitHub Actions automation, and exercises transformation and error handling through tests that validate generated Patient, Encounter, Observation, Practitioner, and AllergyIntolerance resources.
- DETL-design-challenge** 2023
Architectural design for a dynamic ETL system that transforms heterogeneous clinical research data into the OMOP Common Data Model. The design combines detailed source to target specifications, human readable transformation rules, generated SQL, and explicit review, testing, and debugging stages so that common mappings can be automated without removing expert oversight from complex semantic decisions. It also documents data quality controls, terminology mapping, de-identification considerations, and implementation options involving Apache NiFi, Azure Data Factory, and Python.

covid-dataset	2020 - 2022
Azure Data Factory data engineering project that ingests public ECDC COVID-19 and population data over HTTP and Blob Storage, lands raw data in ADLS Gen2, transforms cases, deaths, and hospital admissions with mapping data flows, and publishes curated snapshots to Azure SQL Database. Implements metadata-driven, scheduled, and event-driven ingestion; country, population, and calendar enrichment; idempotent, failure-safe staged SQL publication with audit records and concurrency controls; managed-identity and least-privilege security; ARM deployment templates; and continuous integration checks for secrets, ADF references, T-SQL, and generated artifacts.	
zams_cluster_candidacy	2019 - 2020
Python scientific computing pipeline for evaluating stellar cluster candidacy from Washington system photometry. The analysis filters observations by photometric uncertainty and cluster radius, incorporates decontamination membership probabilities, and corrects color-magnitude diagrams for reddening and distance. It then traces candidate cluster sequences with kernel density contours and compares them with zero age main sequence and isochrone models across age and metallicity ranges, producing per cluster and aggregate diagnostic plots.	
quality_measures_engine	2018 - 2019
Python/PyMongo prototype for defining and evaluating population health quality measures from normalized patient records. It represents measure metadata, parameters, calculated dates, initial populations, denominators, numerators, exceptions, and exclusions as reusable JSON definitions, then translates those definitions into MongoDB queries or map reduce operations. Fixtures and automated tests demonstrate the design using sample NQF cervical cancer screening and pneumonia vaccination measures.	
gamma_rays	2019
Python and Jupyter machine learning research project for reconstructing events from high energy gamma ray telescope data. It reads simulated FITS event tables, balances gamma ray and hadronic background samples, and compares random forest, multilayer perceptron, and exploratory neural network classification approaches using confusion matrices, ROC/AUC, Q-factor, and calibration metrics. The project also trains regression models for event energy and direction reconstruction and includes reusable prediction code, persisted models, technical reports, and disseminated notebook analyses.	
Star-Chart	2015
Browser based HTML, CSS, and JavaScript astronomy application that renders the visible night sky for a selected location and date. It accepts browser geolocation or manually entered coordinates, calculates Julian date and local sidereal time, projects catalog stars according to right ascension, declination, and magnitude, and computes positions for the Sun, Moon, and visible planets. Stars, planets, and optional constellation lines are drawn on an interactive, zoomable canvas.	

Professional Recognition

Special Recognition, Health Data Movers - Client Delivery	Jul. 2025
Recognized for diagnosing undocumented source-key versioning in a revenue-cycle integration and implementing composite business grains with deterministic surrogate keys across 10 charge and payment models, eliminating duplicate-key failures and stabilizing client delivery.	
Arkos Health BI Star	Nov. 2023
Internal Business Intelligence recognition received during a period marked by establishing the team's GitHub workflow, delivering 27 BI reports, creating the Arkos Power BI theme and visual standards, and strengthening PHI deidentification and analytics documentation supporting value-based-care operations.	
Manager Recognition for Transition Leadership, Merative - Systems Migration	Feb. 2022
Recognized by management for transition leadership while assuring the HDFS-to-Azure platform migration: built a seven-layer Python and SQL acceptance suite, validated Azure Data Factory, Blob Storage, and Synapse workflows, and led HIPAA-compliant deidentification testing across data representing approximately 300 million patient lives, reaching sign-off with zero unresolved discrepancies.	
IBM Champion Learner	2016 - 2022
Progressed from Bronze in 2016 to Silver in 2017, then earned Gold annually from 2018 through 2022 through IBM Your Learning continuing education, completing 1,477 learning hours and earning 43 credentials. Gold recognition required at least 200 annual learning hours and three new badges or certifications.	
Lab Services Award, IBM - Cross-Team Delivery Coordination	Jul. 2018
Recognized for leading a multi-organization healthcare integration spanning more than 70 EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules, and coordinating engineering and delivery teams to meet client commitments.	
Eminence and Excellence Award, IBM - Data-Curation Tool Development	Jul. 2017
Recognized for engineering an Apache Pig pipeline that parsed HL7 v2 and v3 data for approximately 30 internal users, replacing a Mirth-based workflow and eliminating \$30,000 in annual vendor costs.	
Color Lab Automation Award, PPG Industries - Expanded Reporting	Sep. 2012
Recognized as part of the Color R&D team supporting PPG's new computer-controlled automotive-refinish color-matching laboratory in Cleveland; expanded automated data collection and analysis reporting used to evaluate pigments and increase formulation throughput, precision, and operational efficiency.	
Mary J. Hanna Research Fellowship, University of Cincinnati	Aug. 2010
Department of Physics fellowship providing financial support for completion of M.S. research using near-infrared photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk star clusters.	

Academic Scholarships and Honors

2000 - 2010

Received University Graduate Scholarships supporting physics study from 2008 to 2010; earned University of Cincinnati Dean's List recognition from 2005 to 2007; received Ohio University's faculty-endorsed Provost's Undergraduate Research Fund grant for the 2003 low-toxic photo-etch project and earned Dean's List recognition from 2002 to 2004; received the Ohio Proficiency Scholarship in 2000 after passing all five sections of the state's twelfth-grade proficiency examination.

Teaching and Academic Research

Across these institutions, used interactive, peer-led, and just-in-time teaching Built an interdisciplinary foundation in scientific research, quantitative analysis, and education before entering the technology industry. Conducted published astrophysics and experimental printmaking research, developed calibrated and reproducible data products, and taught physics and astronomy across laboratory, technical, general education, and online settings. Established a teaching practice centered on evidence, accessibility, interactive learning, and continuous improvement—skills later applied to technical leadership, documentation, mentoring, and stakeholder communication.

Lorain County Community College

Aug. 2016 - May 2019

Adjunct Professor - Taught the laboratory component of algebra-based General Physics I (then PHYC 151) and the three-credit ASTY 152 Solar System Astronomy lecture course.

- Exceeded institutional student-evaluation benchmarks by an average of 16% in 2017 and 17% in 2018.
- Reduced reported student confusion by 28% by reorganizing online instructional materials and reducing the need for clarification during office hours.
- Developed and taught online astronomy instruction covering astronomical observation, planetary-system formation, terrestrial and Jovian planets, small bodies, and space exploration.
- Used Just-in-Time Teaching strategies to adapt instruction for underperforming students based on pre-class responses.
- Developed and published more than 70 hours of publicly available recorded lectures and laboratory content to support flexible, on-demand learning.
- Guided students through laboratory work in mechanics, energy, momentum, rotation, and thermal physics, emphasizing measurement uncertainty, quantitative analysis, and scientific reporting.

Cincinnati State Technical and Community College

Feb. 2011 - Jun. 2011

Adjunct Professor - Developed and delivered the integrated Technical Physics I–III lecture and laboratory sequence for technical-program students with varied mathematics preparation.

- Wrote laboratory instructions and coordinated team-based measurement, data-analysis, and technical-report work.
- Delivered two weekly lecture hours and three laboratory hours per course across applied mechanics, thermodynamics, electricity and circuits, waves, and optics.

Miami University

Aug. 2010 - Jun. 2011

Adjunct Professor - Developed and taught three-credit general-education courses PHY 101 Physics and Society and PHY 111 Astronomy and Space Physics.

- Designed a concept-first applied-physics curriculum for non-science majors, translating advanced technical subjects into accessible, decision-ready frameworks without relying on calculus.
- Structured lessons around consequential real-world questions, teaching students to distinguish scientific constraints from policy and value judgments and to communicate evidence-based conclusions to nontechnical audiences.
- Taught students to use order-of-magnitude estimates, energy density, power, conservation laws, and comparative risk to evaluate claims involving transportation fuels, electricity generation, climate, and emerging technologies.
- Connected nuclear physics and radioactivity with power generation, medical imaging, radiation risk, and nuclear security; linked gravity, waves, and the electromagnetic spectrum with satellites, GPS, radar, communications, and remote sensing.
- Taught astronomy topics spanning planetary systems, stellar evolution, galaxies, cosmology, and the scientific search for extraterrestrial life.

University of Cincinnati

Jan. 2006 - Sep. 2010

Held concurrent research and teaching appointments in physics, astronomy, and engineering computation.

Research Assistant, Department of Physics

Sep. 2006 - Sep. 2010

Conducted near-infrared astronomical research using observations from the 0.8–5.4 micrometer SpeX spectrograph at the NASA Infrared Telescope Facility.

- Combined spectral classifications with 2MASS photometry to estimate cluster extinction ($A_V \sim 21\text{--}23$), distance (1.0–1.75 kpc), age of no more than a few million years, and mass of approximately 10^3 solar masses.
- Helped classify the cluster's massive stellar population, identifying an O5 V dwarf, several early-B stars, and a massive young stellar object exhibiting CO band-head emission from dense circumstellar material.
- Published the resulting 130-source astrometric and photometric catalog through the international VizieR astronomical data service.
- Co-developed an astrometrically calibrated catalog of 130 sources by registering SpeX acquisition images to 2MASS; compiled available *JHK* photometry and used DAOPHOT to separate blended sources in the crowded

field.

- Cross-matched infrared and radio-source positions and identified candidate near-infrared counterparts for three deeply embedded radio sources associated with early star formation.
- Reduced and analyzed 0.8–2.4 micrometer, $R \sim 2000$ SpeX spectra and measured diagnostic H I, He I, He II, C IV, N III, and CO spectral features across nine candidate cluster members.
- Estimated stellar radial velocities by measuring pixel-space shifts of CO absorption features against normalized standard-star spectra.
- Acquired follow-up L-band spectra of Galactic OB stars with SpeX during three consecutive nights at NASA's IRTF.
- Performed astronomical spectroscopy, photometry, classification, calibration, signal analysis, and noise reduction using IRAF, IDL, SpeXtool, Mathematica, and Perl.

Teaching Assistant

Jan. 2006 - Jun. 2010

Led approximately 60 laboratory meetings for more than 100 engineering, science, and preprofessional students in the algebra/trigonometry-based College Physics sequence (PHYS 111–113) and calculus-based General Physics sequence (PHYS 211–213).

- Designed interactive laboratory instruction using Peer Instruction and Interactive Lecture Demonstrations to identify and address student misconceptions in real time.
- Taught experimental design, measurement uncertainty, data acquisition, graphical analysis, and technical reporting using PASCO DataStudio and Microsoft Excel.
- Supported group recitations in freshman engineering physics through demonstrations, short lectures, and individual assistance.
- Served as teaching assistant and grader for the corresponding College Physics lecture sequence (PHYS 101–103) and General Physics lecture sequence (PHYS 201–203).
- Supported Astronomy and the Natural Universe lectures (PHYS 121–122) and astronomy laboratories (PHYS 125–127) through grading and instructional assistance.

Engineering Teaching Assistant

Oct. 2009 - Dec. 2009

Prepared and delivered Mathematica instruction for Numerical Methods for Engineering Design, an upper-division aerospace-engineering course.

- Connected computational methods with aerospace analysis and design problems, showing students how to translate mathematical formulations into executable workflows and interpret the resulting output.
- Reinforced numerical accuracy, error, stability, and method selection so students could evaluate the credibility of computed engineering results.
- Developed step-by-step demonstrations applying symbolic and numerical computing to interpolation and approximation, numerical integration, nonlinear equation solving, systems of equations, and ordinary differential equations.

Ohio University

Oct. 2003 - May 2004

Student Researcher, Fine Arts - Earned a competitive Provost's Undergraduate Research Fund award to adapt a wintgreen-oil toner-transfer process from lithography to intaglio photo-etch printing.

- Replaced xylene- and acetone-based photo-transfer steps with methyl salicylate, eliminated the need for specialized exposure equipment, and demonstrated a process using less than \$20 in materials.
- Developed and tested a complete copper- and zinc-plate workflow spanning surface preparation, asphaltum grounding, high-contrast toner transfer, burnishing, clearing, aquatint, and staged etching.
- Characterized critical process variables and failure modes, including ground thickness and cure time, solvent saturation, image resolution, transfer pressure, evaporation time, and temperature control during aquatint fusion.

Publications and Research Outputs

Applied Research and Technical Writing

Bubnick, Ben. (2025). "Reproducible Postgres load of Tuva seed datasets." Internal applied research series, GrowData Analytics/HSEA. Document tracking no. 20250401-A. Internally published by HSEA, Aug. 2025. Author's version, Apr. 1, 2025.

Bubnick, Ben. (2020). "Surgical Scheduling Optimization and Procedure Duration Prediction." Internal project report, IBM. Document tracking no. RC25712.

Bubnick, Ben. (2017). "Process and Requirements for CCD Data Integration." Internal white paper, IBM. Document tracking no. RC25043.

Bubnick, Ben. (2013). "Characterization of Surface Coatings Using RT Models for Color Matching." Unfiled patent application authored for PPG Industries and retained by the company as a trade secret. Ref. 1284 - February 2013.

Bubnick, Ben. (2010). "The B Stars of Westerlund 1." Unpublished internal stellar-spectroscopy analysis, University of Cincinnati.

Journal Articles and Data Catalogs

Bubnick, Ben. (2015). *AI paint formula prediction from spectroscopic measurements* (Version 1.2) [Interactive web application]. shinyapps.io.

Bubnick, Ben. (2010). *Massive stellar clusters in the disk of the Milky Way Galaxy* [Master's thesis, University of

Cincinnati]. OhioLINK Electronic Theses and Dissertations Center. OhioLINK record

Hanson, M. M., & Bubnick, Ben. (2008). *JHK photometry in cluster [BDS2003] 107* [Data set]. VizieR Online Data Catalog (Catalog J/PASP/120/150). <https://doi.org/10.26093/cds/vizieer.61200150>

Hanson, M. M., & Bubnick, Ben. (2008). Near-infrared spectroscopy of candidate members of the Galactic cluster [BDS2003] 107. *Publications of the Astronomical Society of the Pacific*, 120(864), 150–159. <https://doi.org/10.1086/527403>

Bubnick, Ben. "A Novel Method for Low-Toxic Photo-Etch Printing." *MAPC Journal*, vol. 11, no. 2, Fall/Winter 2003, pp. 12-13.

Speaking Engagements

Surgical Scheduling Optimization and Procedure Duration Prediction - IBM AI Hackathon - Dec. 2020

Presented a prototype that combined historical operating-room utilization and the current surgical wait list to optimize scheduling and improve total procedure-time estimates beyond traditional fixed-ratio methods; outlined CNN/RNN case-card prediction as a path toward a potential new offering.

IBM Watson Care Insights - IBM REACH Initiative - 2019

Delivered conceptual and hands-on training with synthetic patient data, explaining how longitudinal records, real-time HL7 feeds, and FHIR configuration support clinical-data integration and the delivery of insights within physicians' EHR workflows.

Machine Learning with Real-World Health Data - Content and Data Analytics Learning Exchange - 2018

Introduced an implementation-oriented approach to applying machine-learning algorithms to real-world health data, connecting clinical-data preparation, model development, evaluation, and interpretation.

Natural-Language Processing for Unstructured Data Lakes - Innovation Research Interchange Fall Networks Conference - Sep. 2018

Presented a healthcare case study using natural-language processing and topic modeling to turn nearly six million semi-structured provider notes in a data lake into analyzable themes and features.

Hadoop Ingestion and Data-Lake Concepts - Cross-Team Training Series - 2016

Designed and led a seven-part series connecting hands-on Hadoop/HDFS ingestion workflows with distributed storage, transformation, validation, troubleshooting, and the use of data lakes for heterogeneous clinical data.

Massive Stellar Clusters in the Disk of the Milky Way Galaxy - M.S. Astrophysics Thesis Defense, University of Cincinnati - Jun. 2010

Defended research using near-infrared broad-band photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk clusters and estimate their distance, age, and approximate mass.

Astronomical Research Using 2MASS Data - 2MASS Research Poster Session - 2008

Presented results from combining 2MASS *JHK* photometry with IRTF/SpeX observations to characterize candidate members of the young Galactic cluster [BDS2003] 107, including its stellar population, extinction, distance, age, and approximate mass.

A Novel Method for Low-Toxic Photo-Etch Printing - Ohio University Research Poster Session - 2004

Presented Provost's Undergraduate Research Fund-supported work adapting a wintergreen-oil toner-transfer process from lithography to intaglio photo-etch printing, including a low-cost copper- and zinc-plate workflow that replaced xylene- and acetone-based transfer steps and eliminated specialized exposure equipment.

Creative and Visual Arts

Media: Relief printmaking (woodcut and linocut), intaglio (etching, drypoint, photo-print, and aquatint), screenprint, monoprint, monotype, and papermaking; oil and acrylic painting; graphite, pencil, charcoal, ink, marker, pastel, and mixed-media drawing; photography.

Selected Exhibitions

Drawn and Quartered VII, HEDGE Gallery, 78th Street Studios, Cleveland, Ohio May 15–Jun. 5, 2015

Exhibiting Artist - Participated in a group exhibition of drawings. **Works:** *Molly I* (graphite and pastel on paper, 2013; retained); *Molly II* (graphite and pastel on paper, 2013; gifted); *Untitled* (pencil on paper, 2014; retained); *Untitled* (pencil on paper, 2014; retained); *Untitled* (marker on paper, 2015; retained); *Amanda in 6 colors* (pencil on paper, 2014; retained); *Amanda in 3 colors* (pencil on paper, 2014; retained); *Keri* (pencil and pastel on paper, 2015; gifted); *Emily* (pencil and pastel on paper, 2012; retained); *Student at the CMA* (pencil on paper, 2014; retained); *Girl from Treehouse* (pencil and pastel on paper, 2013; retained); *Girl from the Meadow* (pencil and pastel on paper, 2013; retained); *Chatty Nurse on the Bus* (pencil and pastel on paper, 2013; retained).

13th Annual Holiday Store, Heights Arts, Cleveland Heights, Ohio Nov. 6–Dec. 31, 2014

Exhibiting Artist - Presented limited-edition photography and oil paintings in the annual holiday exhibition and sale. **Works:** *Berea Falls* (photograph, 2014; edition of 20, 2 artist's proofs; 7 sold, remainder retained); *Road near Boston* (photograph, 2014; edition of 15, 1 artist's proof; 4 sold, remainder retained); *Lakewood Park* (photograph, 2013; edition of 12, 1 artist's proof; 3 sold, remainder retained); *Brandywine Falls* (photograph, 2013; edition of 25, 2 artist's proofs; 9 sold, 1 gifted, remainder retained); *Rocky River Reservation* (photograph, 2014; edition of 10 sets, 1 artist's-proof set; 2 sets sold, remainder retained); *CMA Gardens* (photograph, 2013; edition of 10, 1 artist's proof; 2 sold, remainder retained); *Cardinal I* (photograph, 2013; edition of 12, 1 artist's proof; 3 sold, remainder retained); *Cardinal II* (photograph, 2014; edition of 20, 2 artist's proofs; 7 sold, remainder retained); *Lake Erie I* (oil on canvas, 2014; retained); *Lake Erie II* (oil on canvas, 2014; sold); *Lake Erie III* (oil on canvas, 2014; retained).

Holiday Off The Wall Members Show and Sale, Zygote Press, Cleveland, Ohio Dec. 6–7, 2014

Exhibiting Artist - Presented relief prints and monoprints in the annual members' exhibition. **Works:** *Elise* (woodcut, 2004; edition of 20, 2 artist's proofs; 2 sold, remainder retained); *Struggle* (woodcut, 2004; edition of 20, 2 artist's proofs; retained); *Girl I* (monoprint, 2013; sold); *Girl II* (monoprint, 2013; retained); *Girl III* (monoprint, 2013; sold); *Girl V* (monoprint, 2013; sold).

Waterloo Arts Fest, Collinwood, Cleveland, Ohio Jun. 29, 2013

Exhibiting Artist and Printmaker - Presented limited-edition prints and pulled an open-edition festival print on site for visitors. **Works:** *Lake Erie* (linocut, 2013; edition of 25, 2 artist's proofs; 4 sold, remainder retained); *Lake Erie, Low Water* (woodcut, 2013; edition of 12, 1 artist's proof; 3 sold, remainder retained); *Breakwall* (drypoint, 2012; edition of 10; 2 sold, remainder retained); *Transit* (etching and aquatint, 2012; edition of 15; 2 sold); *Waterloo (festival print)* (linocut pulled on site, 2013; open edition of roughly 40; given away at the booth).

Tremont ArtWalk, Tremont, Cleveland, Ohio 2013

Exhibiting Artist - Presented relief, intaglio, screenprint, monotype, and charcoal work in a group exhibition. **Works:** *Buckeye Carbon* (woodcut, 2013, first state; edition of 10; 1 gifted, remainder retained); *Viaduct* (etching, 2013; edition of 12, 1 artist's proof; retained); *Night Shift, East 30th* (charcoal on paper, 2013; retained); *Analemma* (screenprint, 2013; edition of 20; retained); *Steam Plant* (monotype, 2013; retained).

Holiday Off-the-Wall Exhibition, Zygote Press, Cleveland, Ohio Dec. 1–2, 2012

Exhibiting Artist - Presented relief prints and monoprints in the annual members' exhibition. **Works:** *Elise* (woodcut, 2004; edition of 20, 2 artist's proofs; retained); *Struggle* (woodcut, 2004; edition of 20, 2 artist's proofs; retained); *Girl I* (monoprint, 2013; retained); *Girl II* (monoprint, 2013; retained); *Girl III* (monoprint, 2013; retained); *Girl IV* (monoprint, 2013; sold); *Girl V* (monoprint, 2013; retained).

Juried Exhibition, Zygote Press, Cleveland, Ohio Jun. 1–Jul. 7, 2012

Exhibiting Artist - Presented photo-print etchings in a juried exhibition. **Works:** *Transit of Venus, 5 June 2012* (photo-print etching, 2012; edition of 15, 2 artist's proofs; retained); *Ephemeris (plate I)* (photo-print etching, 2012; edition of 10, 1 artist's proof; retained).

Neighborhood Artists, Kennedy Heights Arts Center, Cincinnati, Ohio 2010

Exhibiting Artist - Participated in a group exhibition of paintings by neighborhood artists. **Works:** *Montgomery Road, Rain* (oil on panel, 2010; gifted); *Back Lot, Kennedy Heights* (oil on canvas, 2010; gifted); *Basin Fog* (oil on canvas, 2009; gifted); *Ludlow Viaduct* (acrylic on panel, 2009; gifted); *Night Kitchen* (oil on panel, 2010; gifted).

Final Friday Open Studios, Pendleton Art Center, Cincinnati, Ohio 2007–2008

Studio Artist - Presented paintings from a shared studio during recurring open-studio events. **Works:** *Mt. Adams from Columbia Parkway* (oil on canvas, 2007; sold); *Over-the-Rhine Window* (oil on panel, 2007; sold); *Basin Fog (first version)* (oil on canvas, 2007; gifted); *Eden Park Overlook* (oil on panel, 2008; sold); *Pendleton Studio, Afternoon* (oil on panel, 2008; gifted); *Viaduct at Dusk* (acrylic on panel, 2008; gifted).

B.F.A. Thesis Exhibition, Jennings House Gallery, Ohio University, Athens, Ohio May 2004

Exhibiting Artist - Presented senior printmaking work spanning relief, intaglio, and screenprint. **Works:** *Self Portrait I* (woodcut, 2003; edition of 10, 2 artist's proofs; retained); *Self Portrait II* (woodcut, 2004; edition of 10, 2 artist's proofs; retained); *Self Portrait III* (woodcut, 2004; edition of 10, 2 artist's proofs; retained); *Brick Kiln* (etching and aquatint, 2004; edition of 12, 1 artist's proof; retained); *Coal Seam* (aquatint and drypoint, 2004; edition of 8, 1 artist's proof; retained); *Court Street, 2 a.m.* (screenprint, 2004; edition of 20, 2 artist's proofs; retained).

Grand Opening, Lacuna Gallery, Athens, Ohio Apr. 2004

Exhibiting Artist - Displayed an original screenprint in the gallery's opening exhibition. **Work:** *Court Street, 2 a.m.* (screenprint, 2004; edition of 20; retained).

Undergraduate Juried Exhibition, Ohio University Art Galleries, Athens, Ohio 2004

Exhibiting Artist - Presented three prints in the university's juried exhibition. **Works:** *Brick Kiln, Nelsonville* (etching and aquatint, 2004; edition of 12, 1 artist's proof; retained); *Ridge and Hollow II* (woodcut, 2004; edition of 10, 2 artist's proofs; retained); *Court Street, 2 a.m.* (screenprint, 2004; edition of 20, 2 artist's proofs; retained).

Student Print Exchange, Ohio University Art Galleries, Athens, Ohio Feb. 2004

Organizer and Exhibiting Printmaker - Organized a student print exchange, editioned a work to the size of the exchange, and exhibited it alongside the received prints. **Work:** *Company Houses (exchange impression)* (etching, 2004; edition of 22 for the exchange, 2 artist's proofs; 20 distributed to participants, artist's proof retained).

Print Exchanges, Conferences, and Commissions

Mid America Print Council Conference Print City: Detroit 2014, Wayne State University, Detroit, Michigan Sep. 24–27, 2014

Contributed a woodcut to a conference exchange portfolio. **Work:** *Buckeye Carbon (exchange impression)* (woodcut, 2014; edition of 30, 2 artist's proofs; 28 distributed through the portfolio, artist's proof retained).

Southern Graphics Council Conference Making Histories: Revolution and Representation, Boston, Massachusetts Apr. 2–5, 2003

Contributed a linocut to a student exchange portfolio. **Work:** *Night Loading Dock (exchange impression)* (linocut, 2003; edition of 25, 2 artist's proofs; 24 distributed through the student portfolio, artist's proof retained).

Ohio University School of Art Commission 2003

Commissioned Artist - Created an original woodcut commissioned by the School of Art; a commissioned impression was presented to Raymond Tymas-Jones, Dean of the College of Fine Arts. **Work:** *Tree on Paper* (woodcut, 2003; edition of 10, 1 artist's proof; commissioned impression gifted).

Studio Access, Workshops, and Visiting-Artist Projects

Zygote Press, Cleveland, Ohio	2011–2014
Completed independent studio work in etching, relief, and screenprint.	
Morgan Art of Papermaking Conservatory, Cleveland, Ohio	Summer 2012
Completed a papermaking workshop.	
Manifest Drawing Center, Cincinnati, Ohio	2009–2010
Participated in open figure-drawing sessions.	
Art Academy of Cincinnati Community Education	Summer 2005
Completed oil-painting instruction in Cincinnati, Ohio.	
Debra Fisher Visiting-Artist Print Project, Ohio University	Summer 2003
Studio Assistant - Supported plate and chine colle preparation, proofing, and editioning.	
Dan Murphy Visiting-Artist Print Project, Ohio University	Summer 2003
Studio Assistant - Supported screen preparation, proofing, and editioning.	

Volunteer and Community Experience

HSEA	Feb. 2025 - May 2025
ETL Brigade Volunteer - Built a reusable PostgreSQL data model and reproducible loading process for Tuva seed datasets, enabling community-health organizations to prototype analytics using claims, visits, and eligibility data. Added schema setup, normalization, and validation components to make the data consistently loadable and testable.	
CSforAll	Oct. 2018 - May 2019
Instructor - Supported computer science instruction for Cleveland Metropolitan School District high school students as part of a district wide effort to make rigorous computer science coursework available in every high school. Helped broaden access for students from underrepresented and low-income populations, with particular attention to learners without reliable computer or Internet access at home.	
HIT in the CLE	Mar. 2017 - Mar. 2019
Coach - Coached five-student high school teams through a data science competition, helping participants analyze a large dataset, develop evidence-based conclusions, and communicate their findings. Introduced students to health-information-technology careers while building data analysis and presentation skills and supporting a more diverse Northeast Ohio Health IT talent pipeline.	
Code for America	Jan. 2015 - Jun. 2015
ETL Guild Volunteer - Applied ETL and data science methods to initiatives that made difficult to use public records more accessible and machine-readable for research, planning, and civic use. Supported broader efforts to improve government transparency and help residents engage with public information.	
St. Herman House - FOCUS Cleveland, Cleveland, Ohio	Sep. 2013 - Mar. 2014
Daily Meal Program Volunteer - Supported a year-round meal program providing three meals a day to neighbors in need, helping deliver welcoming, dignity centered service.	
Distributed meal tickets, plated food, handed meals to guests, and served water and desserts in the dining room.	
Heights Observer, Cleveland Heights, Ohio	Feb. 2012 - May 2014
Volunteer Editor - Edited resident-submitted reporting for a nonprofit, citizen-produced newspaper serving Cleveland Heights and University Heights, improving accuracy and clarity while preserving each contributor's voice. Also contributed local arts and culture coverage, supporting civic participation, public discourse, and independent community journalism.	
The News Record, University of Cincinnati	Feb. 2008 - Sep. 2010
Editorial Contributor - Wrote and edited campus content for the university's independent, student-run newspaper. Helped inform and educate the university community, amplify student perspectives, and maintain accurate, timely coverage of campus life.	
TCP World Academy	Sep. 2008 - May 2009
Tutor - Tutored students in support of TCP World Academy's mission to develop academically engaged, independent learners. Reinforced classroom learning through technology and participatory micro-society activities that connected students' education with meaningful participation in a global society.	
Project Astro at Ethel M Taylor Academy	Nov. 2008
Volunteer Astronomer - Worked with an educator at Ethel M Taylor Academy to provide hands-on, inquiry based astronomy and physical science activities. Helped students investigate scientific questions and engage with astronomy as practicing scientists through the national Project ASTRO educator-astronomer partnership model.	
Sigma Theta Epsilon, Beta Gamma Chapter, Cincinnati, Ohio	2006 - 2008
Chapter Volunteer - Participated in volunteer service through the University of Cincinnati chapter of an interdenominational Christian fraternity.	
Reach Out on Campus, Ohio University, Athens, Ohio	Sep. 2000 - Jun. 2004
Volunteer - Supported community meal services through the Christian campus ministry during undergraduate study.	
- Routinely served meals and cleaned up at weekly Tuesday community meals at Athens Church of Christ. - Participated in on-campus meal services open to the public.	

Leadership, Memberships, and Activities

Zygote Press, Cleveland, Ohio	2011 - 2014
Member and independent studio artist working in etching, relief, and screenprint. Zygote is a nonprofit community printmaking studio, gallery, archive, and education center that expands artists' access to affordable professional workspace, instruction, exhibitions, residencies, and collaborative development.	

Detroit Color Council 2012 - 2013

Member of a nonprofit industry forum promoting communication and education among engineers, designers, scientists, and other professionals involved in automotive color and appearance. The council advances shared technical knowledge and practices intended to improve the quality of colored and decorative materials.

American Institute of Aeronautics and Astronautics Cincinnati Student Chapter 2009 - 2011

Member of a student chapter connecting participants with the aerospace profession and AIAA's emphasis on innovation, technical excellence, and professional collaboration. Participated in Students for the Exploration and Development of Space in 2009, supporting its mission to advance student involvement in space exploration through educational and engineering projects.

- Vice President, 2010.

University of Cincinnati Astronomy Club 2008 - 2010

Founding member in 2008 who helped establish a student community intended to cultivate interest in astronomy through observation, education, and shared exploration of the night sky.

- Secretary, 2009–2010.
- Treasurer, 2008–2009.

Additional Academic Activities 2006 - 2010

Participated in the Large Synoptic Survey Telescope seminar group, UC SPARKS, and University of Cincinnati physics departmental seminars, which provided forums for research discussion and scientific exchange. Also participated in the University of Cincinnati Mountaineering Club, which promotes safe wilderness skills, outdoor appreciation, stewardship, and community through instruction and group outings.

Society of Physics Students Cincinnati Chapter 2005 - 2008

Helped lead a chapter of the national professional association for physics students, which builds supportive scientific communities and promotes education, research, professional development, outreach, and public understanding of physics.

- Treasurer, 2006–2007.
- Secretary, 2005–2006.

Sigma Theta Epsilon Beta Gamma Chapter 2006 - 2007

Member of an interdenominational Christian service fraternity organized around faith, fellowship, service, and meeting unmet human needs.

- Led planning for the Beta Gamma chapter's 10th-anniversary banquet.
- Chaplain, 2006–2007 academic year.

Mid America Print Council 2003 - 2004

Student member of an educational and community-based organization that supports traditional and contemporary printmaking through conferences, exhibitions, publications, professional exchange, and research. Its stated values emphasize inclusion, diversity, integrity, and creative expression.

Southern Graphics Council (now SGC International) 2002 - 2004

Student member of an educational nonprofit that advances printmaking and graphic arts through professional exchange, exhibitions, publications, conferences, and recognition of artists. The organization promotes research, public appreciation, creative dialogue, and participation across diverse communities.

Athens Wargamers 2001 - 2004

Member of an Ohio University student community centered on tabletop wargaming, strategic problem solving, and social play.

- Treasurer, 2002–2003 academic year.

Reach Out on Campus, Ohio University Sep. 2000 - Jun. 2004

Member of a Christian campus ministry at Ohio University focused on fellowship, spiritual growth, campus outreach, and community service.

- Student Leader, 2003–2004 academic year.

Academic Services**Huntington Learning Center** Feb. 2012 - Aug. 2013

Math/Science Exam Prep Teacher - Provided individualized tutoring and test preparation, specializing in ACT and SAT exams while also supporting students in math, science, and other academic areas.

- Adapted instruction to individual learning needs across math, science, and additional coursework beyond college-entrance exam preparation.
- Taught exam content and test-taking strategies through targeted practice, helping students strengthen subject knowledge, pacing, and confidence for the ACT and SAT.

University of Cincinnati Geology Math Physics Library Mar. 2006 - Sep. 2008

Student Librarian - Supported research and collection operations for a specialized university library.

- Assisted students, faculty, and other patrons with research and reference needs and helped locate relevant materials.
- Maintained Library of Congress-classified collections, conducted inventory, and investigated missing items.

University of Cincinnati Learning Communities Program Jun. 2006 - Aug. 2008

Peer Leader - Facilitated a learning community for approximately 25 first-year students co-enrolled in multiple courses, supporting their academic, social, and professional transition to university life.

- Provided individual and small-group mentoring, helping students form study networks, navigate university

systems, and connect with faculty, advisers, and academic-support services.

- Planned community-building activities and coordinated with faculty, advisers, program staff, and campus organizations to strengthen belonging, peer accountability, and academic engagement.
- Prepared and facilitated twice-weekly meetings on course assignments, study strategies, time management, academic expectations, campus resources, and first-year transition needs.

Private Tutoring

Jun. 2006 - Aug. 2007

Private Tutor - Provided sustained one-on-one instruction to a student completing the calculus-based introductory physics sequence, PHYS 201–203.

- Diagnosed misconceptions through guided questioning and targeted practice, reinforcing difficult concepts with step-by-step explanations and alternative solution strategies.
- Taught a repeatable problem-solving framework built around diagrams, governing physical principles, dimensional analysis, estimation, and interpretation of calculated results.
- Adapted short lectures, demonstrations, and representative worked problems to the student's individual learning needs and progress across the three-course sequence.

University of Cincinnati

Summer 2006

Supplemental Instruction Leader - Supported an intensive summer program in which academically advanced incoming students completed the full calculus-based introductory physics sequence, PHYS 201–203, before beginning the regular academic year.

- Helped students connect concepts across the compressed three-course sequence through diagrams, dimensional analysis, estimation, and physical interpretation of calculated results.
- Planned and led collaborative review sessions for groups of up to 12 students, integrating physics content with transferable learning strategies, addressing misconceptions, and modeling successful student behaviors.
- Delivered focused problem-solving lectures between formal class meetings, working representative problems step by step and demonstrating efficient solution strategies.

Property Operations and Manual Labor

Doodlebug Farms

Sep. 2024 - Present

Family Farm Operator - Establishes and operates a pre-production orchard with 72 American plum and Chickasaw plum trees.

- Manages seasonal tree care, grounds maintenance, supplies and equipment, and physically demanding outdoor work across the developing orchard.
- Plans staged expansion to approximately 216 trees, three times the current planting, while preparing the orchard to transition from establishment into harvest and sales.

Independent Contractor, Cleveland, Ohio

Jun. 2003 - Apr. 2015

Residential Painter and Maintenance Contractor - Completed home-improvement and property-maintenance projects.

- Prepared interior surfaces, applied paint, protected surrounding finishes, and maintained orderly work areas through project completion.

Cleveland Convention Center, Cleveland, Ohio

Aug. 2013 - Jan. 2014

Event Setup Crew - Set up and broke down trade-show spaces at downtown convention venues.

- Assembled booths, laid carpet, installed pipe and drape, and cleared materials after events.

Cleveland-Area Distribution Warehouse

Aug. 2013 - Jan. 2014

Warehouse Fulfillment Worker - Picked and packed orders for shipment at a distribution facility south of Cleveland.

Downtown Cleveland Construction and Redevelopment Sites

Jan. 2012 - Mar. 2012

General Laborer - Performed construction cleanup and material handling on downtown projects.

- Cleared debris and moved materials at job sites around Public Square and on the Flats East Bank redevelopment.

Independent Contractor, Cleveland, Ohio

Summer 2003

Concrete Repair Laborer - Repaired architectural concrete flooring and residential driveways for home-improvement and maintenance projects.

- Prepared damaged areas, placed and finished repair material, and completed job-site cleanup to restore worn concrete surfaces.

Cleveland-Area Metal-Stamping and Parts Supplier

Summer 2003

Production Floor Laborer - Packed finished metal parts and handled scrap on a small industrial production floor.

- Fed scrap bins and kept finished parts moving through packing and material-handling steps.

Brook Park Warehouse, Ohio

Winter 2003

Truck Loader and Unloader - Loaded and unloaded trucks at a warehouse near Cleveland Hopkins International Airport.

Cleveland-Area Scrapyard and Harbor Boneyard

Winter 2003

Scrap Sorting and Yard Laborer - Sorted metal at a scrapyard near the Cuyahoga River and performed manual labor at a harbor boneyard.

Cleveland-Area Residential Construction Sites

Summer 2002

Construction Cleanup Laborer - Cleaned houses under construction in suburban developments including Strongsville and North Royalton.

- Hauled scrap drywall, swept finished houses, and loaded dumpsters with construction debris.

Cleveland-Area Moving, Property Clearance, and Demolition Assignments Summer 2002
General Laborer - Assisted local movers, cleared foreclosed houses, and worked for several weeks on a crew removing an old plant on Cleveland's East Side.

Gund Arena and Jacobs Field, Cleveland, Ohio Summer 2002
Event Setup and Teardown Crew - Prepared venues for concerts and games and cleaned up after large crowds.
 • Moved folding chairs and staging, dismantled event setups, and completed post-event cleanup.

Crown Metal Finishing Summer 2002
Production Laborer - Supported an industrial chrome-plating operation through hands-on material handling and line preparation.
 • Loaded metal components onto production racks for machine-applied chrome plating, arranging parts for consistent processing and efficient line flow.
 • Sustained production pace during repetitive manual work and followed shop-floor handling and safety procedures around plating equipment.

Gravity Games Summer 2002
Event Operations Crew - Supported venue sanitation and waste-handling operations during a large public sporting event.
 • Collected and disposed of waste and recycling throughout the venue and removed litter from high-traffic areas during and after the event.

FedEx Winter 2002
Package Handler and Loader - Performed time-sensitive loading and material-handling work in a high-volume logistics environment.
 • Completed physically demanding assignments involving repetitive lifting, rapid material movement, and coordination with other handlers.

Sales and Customer Service

Target Winter 2010
Sales Associate - Processed customer purchases and cash and credit transactions in a high-volume retail environment.
 • Assisted customers, stocked merchandise, supported inventory activities, and helped resolve transaction concerns.

FranklinCovey Jun. 2005 - Apr. 2006
Productivity Consultant - Advised customers on planning products, workshops, and productivity needs using detailed product knowledge.
 • Matched products and services to individual customer goals and supported cash and credit transactions.

Dillard's Aug. 2004 - Jun. 2005
Sales Associate - Helped customers evaluate merchandise and complete purchases.
 • Supported merchandising, stocking, opening and closing activities, and customer-service issue resolution.

DialAmerica Marketing Winter 2002
Marketing Representative - Supported telephone-based customer outreach and sales operations.
 • Conducted customer outreach and sales conversations by telephone.

Food Service and Hospitality

Lincoln Park Pub Aug. 2013 - Mar. 2014
Cook - Supported restaurant kitchen operations in a fast-paced service environment.
 • Prepared food and coordinated kitchen work to support accurate, timely meal service.

Aladdin's Eatery Summer 2002
Dishwasher - Supported dish-room sanitation and restaurant kitchen operations.
 • Maintained clean dishes, utensils, equipment, and work areas throughout restaurant service.

Barnacle Bill's Lobster House Summer 2001
Cook - Supported kitchen production in a fast-paced restaurant environment.
 • Prepared food and helped maintain timely meal production during restaurant service.

Cravings Cafe Winter 2001
Barista - Provided beverage preparation and counter service in a cafe environment.
 • Prepared coffee and espresso beverages, served customers, replenished supplies, and maintained the service area.

Pier W Summer 2000
Cook - Supported food preparation and restaurant kitchen operations.
 • Assisted with food preparation, kitchen cleaning, and meal-service activities.

Take 5 Coffee Company Winter 2000
Barista - Supported day-to-day coffee-shop and counter-service operations.
 • Prepared coffee and espresso beverages, stocked supplies, and completed opening and closing activities.

Credentials and Continuing Education

Databricks Academy Credentials Sep. 2026
 Earned 13 Databricks credentials spanning data engineering and data governance.
 • **Data engineering:** Automated Deployment with Declarative Automation Bundles (Credential ID 194935119); Databricks Performance Optimization (Credential ID 194934527); Databricks Data Privacy (Credential ID

194933863); Advanced Techniques with Spark Declarative Pipeline (Credential ID 194933166); Build Data Pipelines with Lakeflow Spark Declarative Pipelines (Credential ID 194932264); Deploy Workloads with Lakeflow Jobs (Credential ID 194931356); Data Ingestion with Lakeflow Connect (Credential ID 194928030); and DevOps Essentials for Data Engineering (Credential ID 194930163).

- **Data governance:** Data Interoperability with Unity Catalog (Credential ID 194981221); Data Governance at Scale (Credential ID 194980190); Catalog Management and Data Organization (Credential ID 194979341); Get Started with Data Governance on Databricks (Credential ID 194978447); and Academy Accreditation - Data Governance Fundamentals (Credential ID 194977782).

Salesforce Trailhead Coursework Completed at Arkos Health

2023 - 2025

Reached Trailblazer Mountaineer rank with 18,725 points and 40 badges.

- **Tableau and data literacy:** Tableau Desktop, Tableau Cloud, Tableau Prep, visualization, data structures and field types, filtering and sorting, aggregation and granularity, data connections, collaboration, and presentation.
- **Analytics, AI, and security:** CRM Analytics, correlation and regression, data-analytics fundamentals, natural-language processing, generative and responsible AI, Agentforce, and data security.
- **Salesforce administration and CRM:** Salesforce CRM, user management and engagement, Lightning Experience, formulas and validations, data modeling and management, and reports and dashboards.

Fundamentals of Visualization with Tableau - University of California, Davis

Jun. 2023

Credential ID M3VVMS4BTXWU.

IBM Data Science Professional Certificate - IBM

2022

IBM Digital Credentials and Coursework

Nov. 2017 - Feb. 2022

Completed 43 distinct IBM credentials spanning data science, machine learning, data engineering, cloud, product management, design thinking, security, blockchain, and quantum computing.

- **Data engineering and big data:** Hadoop Foundations and Programming; Hadoop Data Access; Spark; Apache Kafka data pipelines; Big Data Foundations, Levels 1–2; Data Science Tools.
- **Data science and machine learning:** IBM AI Associate Data Scientist; Data Science Foundations, Levels 1–2; Data Science Methodologies; Python, R, Scala, TensorFlow, GPU-accelerated deep learning, data visualization, and machine-learning coursework.
- **Cloud and security:** IBM Cloud Private Infrastructure and Architecture; IBM Cloud Private Foundation Technology; Cloud Core; Cloud Service Management and Operations; Think Like a Hacker; Watson and Cloud Foundations.
- **Product, Agile, and design:** Agile Explorer; AgileTA Explorer; Product Manager curriculum covering product definition, delivery, business management, and scaling; Enterprise Design Thinking Practitioner and Co-Creator.
- **Emerging technologies:** Blockchain Essentials and Blockchain Essentials V2; Quantum Conversations; Watson Visual Recognition; Cognitive Practitioner.

Certificate of Specialization in Data Science - Johns Hopkins University

2015

Additional Coursework

2012 - 2014

Windows Forms; quantitative finance; information risk; computer networks; computational data analysis; AI planning.

Education

M.S. Astrophysics, University of Cincinnati

Dec. 2010

Completed the thesis *Massive Stellar Clusters in the Disk of the Milky Way Galaxy*, using near-infrared photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk star clusters.

B.F.A. Printmaking, Ohio University

Aug. 2004

Studio focus in intaglio, relief, and screenprint. Completed a senior thesis exhibition and faculty-endorsed Provost's Undergraduate Research Fund work adapting a low-toxic toner-transfer process to intaglio photo-etch printing.